PHYSICS

1. A particle starts oscillating simple harmonically from its equilibrium position with time period T. The ratio of kinetic energy and potential energy of the particle at time t = T/12 is
(1) 1 : 2 (2) 2 : 1 (3) 3 : 1 (4) 4 : 1

2. If the displacement x and velocity υ of a particle executing simple harmonic motion are related through the expression $4\upsilon^2 = 25 - x^2$, then its time period is
(1) π (2) 2π (3) 4π (4) 6π

3. The kinetic energy of a particle executing SHM will be equal to $(1/8)^{th}$ of its potential energy when its displacement from the mean position is(where A is the amplitude)
(1) $A\sqrt{2}$ (2) $\frac{A}{2}$ (3) $\frac{2\sqrt{2}}{3}A$ (4) $A\sqrt{\frac{2}{3}}$

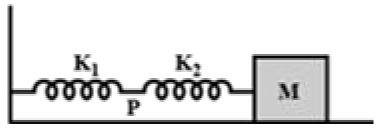
4. Two particles A and B execute simple harmonic motion with periods of T and $\frac{5T}{4}$ respectively. They start simultaneously from mean position. The phase difference between them when A completes one oscillation will be-
(1) 0 (2) $\frac{\pi}{2}$ (3) $\frac{\pi}{4}$ (4) $\frac{2\pi}{5}$

5. The maximum velocity of a particle executing simple harmonic motion with an amplitude 7 mm is 4.4 m/s. The time period of oscillation is
(1) 100 s (2) 0.01 s (3) 10 s (4) 0.1 s

6. The equation of motion of two particles executing S.H.M. are
$y_1 = 10\sin\left(10\pi t + \frac{\pi}{3}\right)m, y_2 = 10\cos\left(8\pi t + \frac{\pi}{4}\right)$
The phase difference between these particles at t = 0.5 s is
(1) $\frac{7\pi}{12}$ (2) $\frac{13\pi}{12}$ (3) $\frac{25\pi}{12}$ (4) $\frac{\pi}{12}$

7. Two simple harmonic motions are represented by the equations $x_1 = 10\sin\left(3\pi t + \frac{\pi}{4}\right)$ and $x_2 = 5(\sin 3\pi t + \sqrt{3}\cos 3\pi t)$. Their amplitudes are in the ratio of
(1) 1 : 2 (2) 1 : 1 (3) 2 : 3 (4) 2 : 1

8. The mass M shown in the figure oscillates in simple harmonic motion with amplitude A. The amplitude of point P is



(1) $\frac{k_1A}{k_2}$ (2) $\frac{k_2A}{k_1}$
(3) $\frac{k_1A}{k_1+k_2}$ (4) $\frac{k_2A}{k_1+k_2}$

9. Two springs of spring constants k_1 and k_2 have equal maximum velocities when executing simple harmonic motion. The ratio of their amplitudes (masses are equal will be)
(1) $\frac{k_1}{k_2}$ (2) $\left(\frac{k_1}{k_2}\right)^{1/2}$ (3) $\frac{k_2}{k_1}$ (4) $\left(\frac{k_2}{k_1}\right)^{1/2}$

10. The bob of a simple pendulum executes SHM in water with a period T, while the period of oscillation of the bob is T_0 in air. Neglecting frictional force of water and given that the density of the bob is $\left(\frac{4}{3}\right) \times 1000\ kgm^{-3}$. The relationship between T and T_0 is
(1) $T = T_0$ (2) $T = 4T_0$
(3) $T = 2T_0$ (4) $T = \frac{T_0}{2}$

11. A sound wave travelling through a medium of bulk modulus B is represented as $y(x, t) = A\sin(kx - \omega t)$ where symbols have their usual meanings. Then, the corresponding pressure amplitude is
(1) BAk (2) $B(A/k)^{1/2}$
(3) B (4) $B(Ak)^{1/2}$

12. A string of linear density 0.2 kg/m is stretched with a force of 500 N. A transverse wave of length 4.0 m is set up along it. The speed of wave is
(1) 50 m/s (2) 75 m/s
(3) 150 m/s (4) 200 m/s

13. A uniform rope of mass 0.1 kg and length 2.45 m hangs from a ceiling. The time taken by a transverse wave to travel the full length of the rope is ($g = 9.8\ m/s^2$)
(1) 1 s (2) 2 s (3) 3 s (4) 4 s

14. Two sound wave travel in the same direction in a medium. The amplitude of each wave is A and the phase difference between the two waves is 120°. The resultant amplitude will be

(1) $\sqrt{2}$ A (2) 2A (3) 3A (4) A

15. **Assertion:** The percentage change in time period is 1.5%, if the length of simple pendulum increases by 3%.

Reason : Time period is directly proportional to length of pendulum.

(1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
(2) If both assertion and reason are true but reason is not the correct explanation of the assertion.
(3) If assertion is true but reason is false.
(4) If the assertion and reason both are false.

16. The stationary wave y = 2a sinkx cosωt in a closed organ pipe is the result of the superposition of y_1 = asin(ωt – kx) and

(1) y_2 = – a cos(ωt – kx)
(2) y_2 = – a sin(ωt + kx)
(3) y_2 = a sin(ωt – kx)
(4) y_2 = a cos(ωt + kx)

17. A closed organ pipe and an open pipe of same length produce 4 beats when they are set into vibrations simultaneously. If the length of each of them were twice their initial lengths, the number of beats produced will be [Assume same mode of vibration in both cases]

(1) 2 (2) 4 (3) 1 (4) 8

18. For a wave displacement amplitude is 10^{-8} m, density of air 1.3 kg m^{-3}, velocity in air 340 ms^{-1} and frequency is 2000 Hz. The intensity of wave is

(1) 5.3×10^{-4} Wm^{-2} (2) 5.3×10^{-6} Wm^{-2}
(3) 3.5×10^{-8} Wm^{-2} (4) 3.5×10^{-6} Wm^{-2}

19. If the length of a stretched string is shortened by 40% and the tension is increased by 44%, then the ratio of the final and initial fundamental frequencies is

(1) 3 : 4 (2) 4 : 3 (3) 1 : 3 (4) 2 : 1

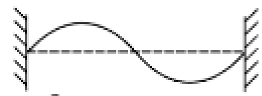
20. A wall clock uses a vertical spring-mass system to measure the time. Each time the mass reaches an extreme position, the clock advances by a second. The clock gives correct time at the equator. If the clock is taken to the poles it will

(1) run slow (2) run fast
(3) stop working (4) give correct time

21. String A has length L, radius of cross-section r, density of material ρ and is under tension T. String B has all these quantities double those of string A. If υ_A and υ_B are the corresponding fundamental frequencies of the vibrating string, then

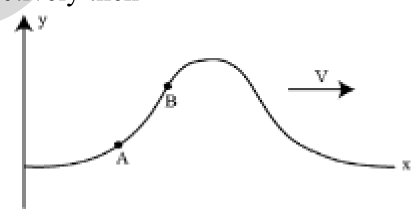
(1) $\upsilon_A = 2\upsilon_B$ (2) $\upsilon_A = 4\upsilon_B$
(3) $\upsilon_B = 4\upsilon_A$ (4) $\upsilon_A = \upsilon_B$

22. The shape of a string on which standing waves are produce at t = 0 is shown here. The suitable equation of standing wave can be



(1) y = A sin kx cos ωt
(2) y = A sin kx sin ωt
(3) y = A cos kx sin ωt
(4) y = A cos kx cos ωt

23. A wave pulse is generated in string that lies along x – axis. At the points A and B, as shown in figure, if R_A and R_B are ratio of wave speed to the particle speed respectively then



(1) $R_A > R_B$
(2) $R_B > R_A$
(3) $R_A = R_B$
(4) Information is not sufficient to decide

24. The potential energy of a particle of mass 1 kg in motion along the x – axis is given by: $U = 4(1 - \cos 2x)$J, where x is in metres. The period of small oscillations (in sec) is:

(1) 2π (2) π
(3) $\frac{\pi}{2}$ (4) $\sqrt{2}\pi$

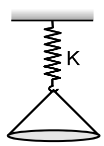
25. The period of small oscillation of a simple pendulum is T. The ratio of density of liquid to the density of material of the bob is ρ(ρ < 1). When immersed in the liquid, the time period of small oscillation will now be

(1) T (2) T(1 – ρ)
(3) $\frac{T}{\sqrt{1-\rho}}$ (4) $T\sqrt{1-\rho}$

26. Motion of an oscillating liquid column in a U-tube is
(1) Periodic but not simple harmonic
(2) Non-periodic
(3) Simple harmonic and time period is independent of the density of the liquid
(4) Simple harmonic and time period is directly proportional to the density of the liquid

27. The speed (v) of a particle moving along a straight line, when it is at a distance (x) from a fixed point on the lines is given by $v^2 = 144 - 9x^2$. Which of the following statement(s) is/are correct?
(1) The magnitude of acceleration at a distance 3 units from the fixed point is 27 units.
(2) The motion is simple harmonic with $T = \frac{2\pi}{3}$ units.
(3) The maximum displacement from the fixed point is 4 units.
(4) All of the above

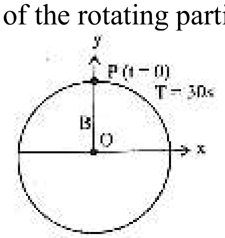
28. A solid ball of mass m is made to fall from a height H on a pan suspended through a spring of spring constant K as shown in figure. If the ball does not rebound and the pan is massless, then amplitude of oscillation is



(1) $\frac{mg}{K}$
(2) $\frac{mg}{k}\left(1 + \frac{2HK}{mg}\right)^{1/2}$
(3) $\frac{mg}{K} + \left(\frac{2HK}{mg}\right)^{1/2}$
(4) $\frac{mg}{K}\left[1 + \left(1 + \frac{2HK}{mg}\right)^{1/2}\right]$

29. A point mass is subjected to two simultaneous sinusoidal displacements in x-direction, $x_1(t) = A \sin \omega t$ and $x_2(t) = A \sin\left(\omega t + \frac{2\pi}{3}\right)$. Adding a third sinusoidal displacement $x_3(t) = B \sin(\omega t + \phi)$ brings the mass to a complete rest. The values of B and ϕ are
(1) $\sqrt{2}A, \frac{3\pi}{4}$
(2) $A, \frac{4\pi}{3}$
(3) $\sqrt{3}A, \frac{5\pi}{6}$
(4) $A, \frac{\pi}{3}$

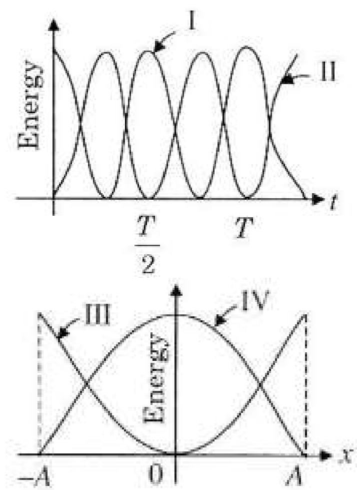
30. Figure shows the circular motion of a particle. The radius of the circle, the period, sense of revolution and the initial position are indicated on the figure. The simple harmonic motion of the x-projection of the radius vector of the rotating particle P is:



(1) $x(t) = B \sin\left(\frac{2\pi t}{30}\right)$
(2) $x(t) = B \cos\left(\frac{\pi t}{15}\right)$
(3) $x(t) = B \sin\left(\frac{\pi t}{15} + \frac{\pi}{2}\right)$
(4) $x(t) = B \cos\left(\frac{\pi t}{15} + \frac{\pi}{2}\right)$

31. In this question, a statement of assertion (A) is followed by a statement of reason (R). Mark the correct choice as:
Assertion: If a pendulum is suspended in a lift and lift is falling freely, then its time period becomes infinite.
Reason: Free falling body has acceleration equal to acceleration due to gravity.
(1) If both assertion and reason are true and reason is the correct explanation of assertion.
(2) If both assertion and reason are true but reason is not the correct explanation of assertion.
(3) If assertion is true but reason is false.
(4) If both assertion and reason are false.

32. The displacement of a linear harmonic oscillator is given by x = A cos ωt. The curves showing the variation of the potential energy with t and x (see figure) are displayed respectively by:



(1) I and III
(2) I and IV
(3) II and III
(4) II and IV

33. A particle of mass m is allowed to oscillate near the point of minima of a vertical parabolic path having the equation $x^2 = 4ay$, where x - axis is along the horizontal direction and y - axis is along the vertical direction. The angular frequency of small oscillations of the particle is
(1) $\sqrt{\frac{8g}{a}}$
(2) $\sqrt{\frac{2g}{a}}$
(3) $\sqrt{\frac{g}{a}}$
(4) $\sqrt{\frac{g}{2a}}$

34. Which one of the following statements is true for the speed'v' and the acceleration 'a' of a particle executing simple harmonic motion?
(1) Value of 'a' is zero, whatever may be the value of'v'.
(2) When 'v' is zero, 'a' is zero.
(3) When'v' is maximum, 'a' is zero.
(4) When'v' is maximum, 'a' is maximum.

35. Two identical sinusoidal waves each of amplitude 10 mm with a phase difference of 120° are travelling in the same direction in a string. The amplitude of the resultant wave is
(1) 5 mm (2) 10 mm
(3) 15 mm (4) 20 mm

36. A transverse harmonic wave on a string is described by, $y(x, t) = 3\sin\left(36t + 0.018x + \frac{\pi}{4}\right)$ where x and y are in cm and t is in s.
Which of the following statements is incorrect?
(1) The wave is travelling in negative x-direction
(2) The amplitude of the wave is 3 cm.
(3) The speed of the wave is 20 m s^{-1}
(4) The frequency of the wave is $\frac{9}{\pi}$ Hz.

37. Equation of a plane progressive wave is given by $y = 0.6.\sin 2\pi\left(t - \frac{x}{2}\right)$. On reflection from a denser medium its amplitude becomes $\frac{2}{3}$ of the amplitude of the incident wave. The equation of the reflected wave is.
(1) $y = 0.6\sin 2\pi\left(t + \frac{x}{2}\right)$
(2) $y = -0.4\sin 2\pi\left(t + \frac{x}{2}\right)$
(3) $y = 0.4\sin 2\pi\left(t + \frac{x}{2}\right)$
(4) $y = -0.4\sin 2\pi\left(t - \frac{x}{2}\right)$

38. A closed pipe of length L contains gas of density ρ_1, another open pipe contains gas at density ρ_2. Both the gases have same compressibility factor and both pipes resonates with same frequency in their first overtone then the length of second pipe is:
(1) $\frac{4L}{3}\sqrt{\frac{\rho_2}{\rho_1}}$ (2) $\frac{4L}{3}\sqrt{\frac{\rho_1}{\rho_2}}$
(3) $\frac{4L}{3}$ (4) $\frac{L}{2}$

39. A wire is 4 m long and has a mass 0.2 kg. The wire is kept horizontally. A transverse pulse is generated by plucking one end of the taut (tight) wire. The pulse makes four trips back and forth along the cord in 0.8 sec. The tension in the cord will be-
(1) 80 N (2) 160 N
(3) 240 N (4) 320 N

40. In this question, a statement of assertion (A) is followed by a statement of reason (R).
Mark the correct choice as:
Assertion : The fundamental frequency of an open organ pipe increases as the temperature is increased.
Reason : This is because as the temperature increases, the velocity of sound increases more rapidly than length of the pipe.
(1) If both assertion and reason are true and reason is the correct explanation of assertion.
(2) If both assertion and reason are true but reason is not the correct explanation of assertion.
(3) If assertion is true but reason is false.
(4) If both assertion and reason are false.

41. In this question, a statement of assertion (A) is followed by a statement of reason (R).
Mark the correct choice as:
Assertion: In a string wave, during reflection from fix boundary, the reflected waves is inverted.
Reason: The force on string by clamp is in downward direction while string is pulling the clamp in upward direction.
(1) If both assertion and reason are true and reason is the correct explanation of assertion.
(2) If both assertion and reason are true but reason is not the correct explanation of assertion.
(3) If assertion is true but reason is false.
(4) If both assertion and reason are false.

42. The equations of displacement of two waves are given as $y_1 = 10\sin\left(3\pi t + \frac{\pi}{3}\right)$
$y_2 = 5(\sin 5\pi t + \sqrt{3}\cos 5\pi t)$
Then, what is the ratio of their amplitudes?
(1) $1:2$ (2) $2:1$
(3) $1:1$ (4) None of these

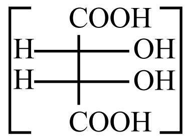
43. Two waves are propagating to the point P by two sources A and B of equal frequency. The amplitude of every wave at P is a and the phase of A is ahead by $\frac{\pi}{3}$ than that of B and the distance AP is greater than BP by 50 cm, the resultant amplitude at the point P will be, if the wavelength is 1 m, is
(1) 2a (2) $a\sqrt{3}$ (3) $a\sqrt{2}$ (4) a

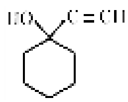
44. Beats are produced by frequencies f_1 and $f_2 (f_1 > f_2)$. The duration of time between two successive maxima or minima is equal to
(1) $\frac{1}{f_1+f_2}$ (2) $\frac{2}{f_1-f_2}$ (3) $\frac{2}{f_1+f_2}$ (4) $\frac{1}{f_1-f_2}$

45. Change in temperature of the medium changes:
(1) Frequency of sound waves
(2) Amplitude of sound waves
(3) Wavelength of sound waves
(4) Loudness of sound waves

CHEMISTRY

46. Increasing order of acid strength among p-methoxyphenol, p-methyl phenol, and p-nitrophenol is -
(1) p-nitrophenol, p-methoxyphenol, p-methyl phenol
(2) p-methyl phenol, p-methoxyphenol, p-nitrophenol
(3) p-nitrophenol, p-methyl phenol, p-methoxyphenol
(4) p-methoxyphenol, p-methyl phenol, p-nitrophenol

47. Meso–tartaric acid  is optically inactive due to the presence of :-
(1) Molecular symmetry
(2) Molecular asymmetry
(3) External compensation
(4) Two asymmetric carbon atoms

48. The IUPAC name of  is -
(1) 1-Ethynyl-1-hydroxycyclohexane
(2) 1-(Hydroxy cyclohexyl) ethyne
(3) 1-Ethynylcyclohexanol
(4) 1-Acetylenyl-1-hydroxycyclohexane

49. **Assertion:** – The melting point of fumaric acid is higher than that of maleic acid.
Reason: – The molecule of fumaric acid are more symmetric than those of maleic acid and hence it gets closely arranged in the crystal lattice.
(1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) If Assertion is True but the Reason is False.
(4) If both Assertion & Reasons are false.

50. Match list I with list II and then select the correct answer from the codes given below the lists

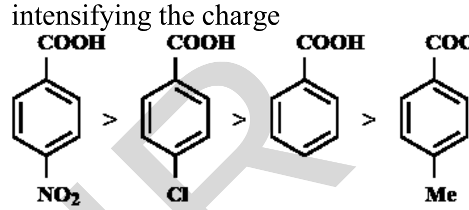
List I		List II	
(A)	Ethyl formate	(i)	Chain isomer of isobutylene
(B)	Ethanamide	(ii)	Higher homologue of formamide
(C)	Propanenitrile	(iii)	Metamer of CH_3–CO–OCH_3
(D)	2-Butene	(iv)	Functional group isomer of ethyl isocyanide

Codes are:

	A	B	C	D
(1)	iii	i	ii	iv
(2)	iii	ii	iv	i
(3)	iii	iv	ii	i
(4)	iii	ii	i	iv

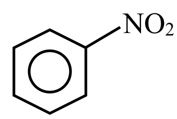
51. Which of the following will show geometrical isomerism
(1) $CH_3CH = CH\ CH_3$
(2) $(CH_3) = C(CH_3)_2$
(3) $(CH_3)_2\ C = CH_2$
(4) $CH_3 – CH = C\ (CH_3)_2$

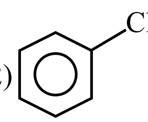
52. **Assertion:** The order of acidic character of the given compounds
Reason: –I and –R effect of the group stabilizes the conjugate base decreasing the –ve charge, whereas +I effect of the group destabilises the conjugate base by intensifying the charge

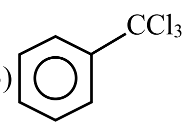


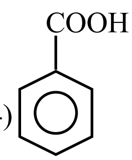
(1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) If Assertion is True but the Reason is False.
(4) If both Assertion & Reasons are false

53. In which of the following compound the electrophile will attack on ortho and para positions:

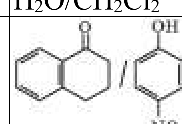
(1) 

(2) 

(3) 

(4) 

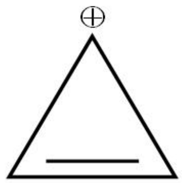
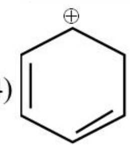
54. Match List -I with List-II

List-I (Mixture of compound)		List-II (Separation Technique)	
A.	H_2O/CH_2Cl_2	I.	Crystallization
B.		II.	Differential solvent extraction
C.	Kerosene/ Naphthalene	III.	Column chromatography
D.	$C_6H_{12}O_6$/NaCl	IV.	Fractional Distillation

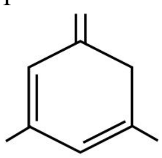
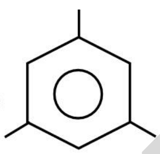
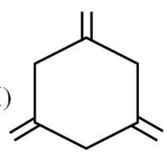
(1) A-III, B-IV, C-II, D-I
(2) A-I, B-III, C-II, D-IV
(3) A-II, B-IV, C-I, D-III
(4) A-II, B-III, C-IV, D-I

55. Given below are two statements:
Statement I: Sulphanilic acid gives esterification test for carboxyl group.
Statement II: Sulphanilic acid gives red colour in Lassigne's test for extra element detection.
In the light of the above statement, choose the most appropriate answer from the option gives below:
(1) Both Statement I and Statement II are correct
(2) Both Statement I and Statement II are incorrect
(3) Statement I is incorrect but Statement II is correct
(4) Statement I is correct but Statement II is incorrect

56. The compound with highest resonance energy is

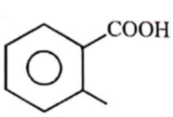
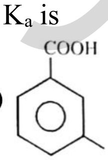
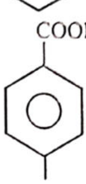
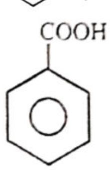
(1) 
(2) 
(3) 
(4) 

57. The correct order of stability for the following compound is

(I) 
(II) 
(III) 

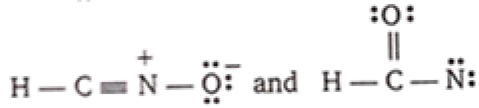
(1) I > II > III (2) II > III > I
(3) II > I > III (4) I > III > II

58. The decreasing order of K_a is

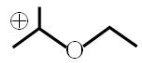
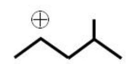
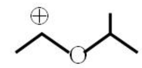
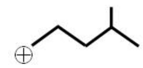
(I) 
(II) 
(III) 
(IV) 

(1) I > II > III > IV (2) IV > II > III > I
(3) I > IV > II > III (4) II > III > I > IV

59. Among the given sets, which represents the resonating structures?

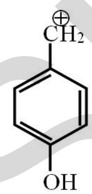
(1) 
(2) 
(3) 
(4) 

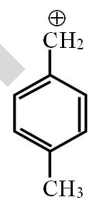
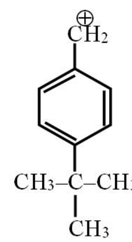
60. The correct stability order for the following species is


(I)

(II)

(III)

(IV)

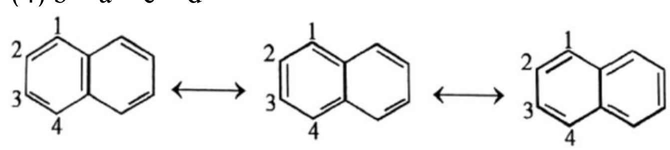
(1) II > IV > I > III (2) I > II > III > IV
(3) II > I > IV > III (4) I > III > II > IV

61.


(a)

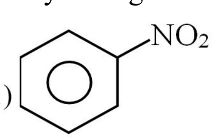
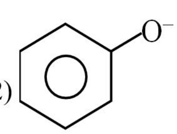
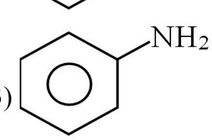
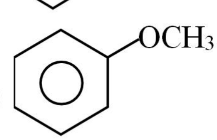
(b)

(c)

(d)

Correct order of carbocation stability is:
(1) b > a > d > c
(2) a > b > d > c
(3) c > d > b > a
(4) b > a > c > d

62. 

These are three canonical structures of naphthalene. Examine them and find correct statement among the following:
(1) All C – C bonds are of same length
(2) $C_1 – C_2$ bond is shorter than $C_2 – C_3$ bond.
(3) $C_1 – C_2$ bond is longer than $C_2 – C_3$ bond
(4) None

63. In which of the following molecules π–electron density in ring is maximum:

(1) 
(2) 
(3) 
(4) 

64. Chloroethane reacts with Na in Presence of dry ether. The Product is
(1) Ethane (2) Propane
(3) Butane (4) Ethene

65. The highest boiling point is expected for
(1) n-butane (2) n-pentane
(3) iso-Pentane (4) neo-pentane

66. When butane 2-ol is heated with H_2SO_4 the major product is:
(1) But-1-en (2) But-2-en
(3) 2-methyl propene (4) Butta-1, 3-diene

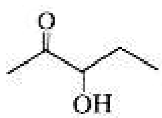
67. A reagent used to test for unsataration of alkene is
(1) Conc. H_2SO_4
(2) Ammonicals Cu_2Cl_2
(3) Ammonicals $AgNO_3$
(4) Solution Br_2 in CCl_4

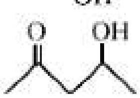
68. Which one of the following reaction is not possible?
(1) $CH_3COONa + HCI \rightarrow CH_3COOH + NaCI$
(2) $CH_3–SO_3H + H–C = C–Na\ CH_3SONa + H–C = C–H$
(3) $R–C = C – H + PhONa \rightarrow PhOH + R – C = C – Na$
(4) $OH – C = C – H + NaH_2 – H – C = C–Na + NH_3$

69. Bromination of (E)-2-butenedioic acid gives.
(1) (2R, 3S)-2, 3-dibromosuccinic acid
(2) (2R, 3R)-2, 3-dibroniosuccinic acid
(3) A mixture of (2R, 3R) and (2S, 3S)-2, 3-dibromosuccinic acid
(3) (2S, 3S)-2, 3-dibromosuccinic acid

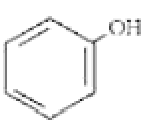
70. Consider the following statements:
(i) Alkene is more reactive than alkyne for electrophilic addition reaction.
(ii) Alkyne gives nucleophilic as well as electrophilic addition reaction.
(iii) Alkyne is more reactive than alkene for nucleophilic addition reaction.
(iv) For electrophilic addition reaction, RI of alkyne is alkyl carbonation.
of these, correct statements are:
(1) only (i) (2) (i) and (ii)
(3) (i), (ii) and (iv) (4) (i), (ii) and (iii)

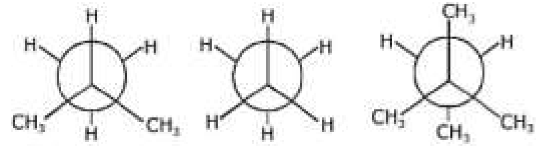
71. Which is more easily dehydrated?

(1) 

(2) 

(3) 

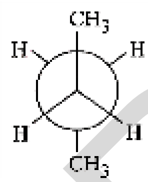
(4) 

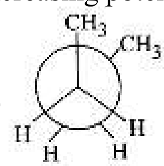
72. 

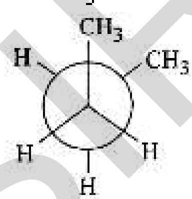
(I) (II) (III)

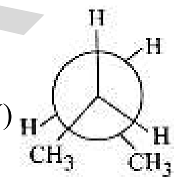
Correct order of mono bromination of given compound is:
(1) I > III > II (2) I > II > III
(3) II > I > III (4) III > II > I

73. Arrange the following conformational isomers of n-butane in order of their increasing potential energy.

(I) 

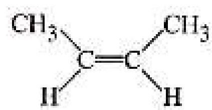
(II) 

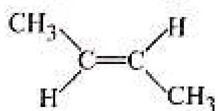
(III) 

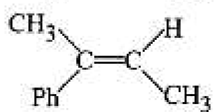
(IV) 

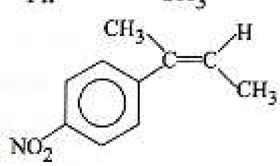
(1) I < III < IV < II (2) I < IV < III < II
(3) II < IV < III < I (4) II < III < IV < I

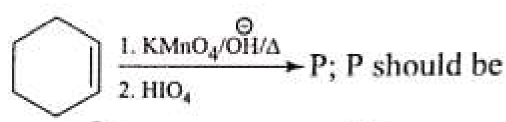
74. The addition of HBr to which of the following alkenes will be most reactive?

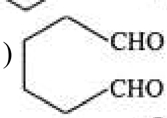
(1) 

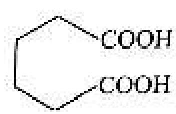
(2) 

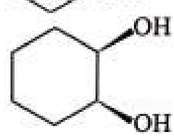
(3) 

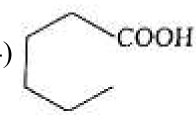
(4) 

75. 

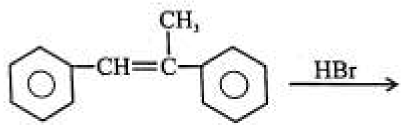
(1) 

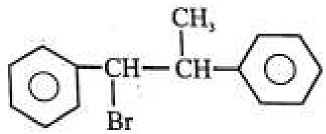
(2) 

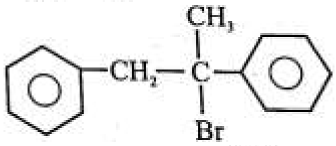
(3) 

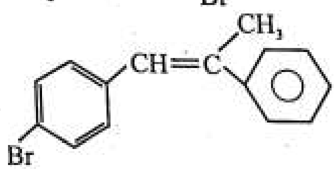
(4) 

76. Which of the following is/are the major product, of the following reaction?



(1) 

(2) 

(3) 

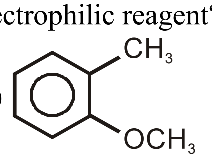
(4) None of these

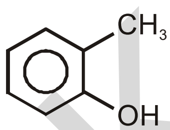
77. Weight (g) of two moles of the organic compound, which is obtained by heating sodium ethanoate with sodium hydroxide in presence of calcium oxide is:

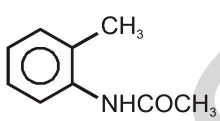
(1) 18 (2) 16

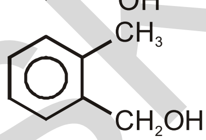
(3) 32 (4) 30

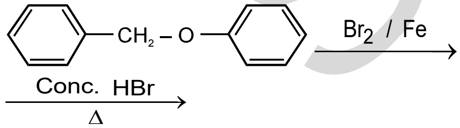
78. Which one of the following is most reactive towards electrophilic reagent?

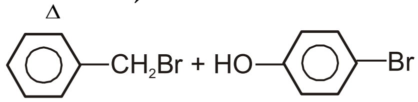
(1) 

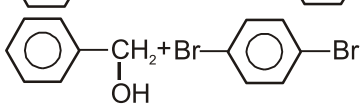
(2) 

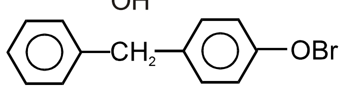
(3) 

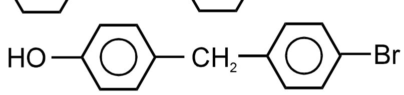
(4) 

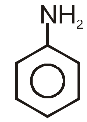
79. 

(1) 

(2) 

(3) 

(4) 

80. Electrophilic substitution of  with bromine water gives :

(1) 2,3,4-Tribromo aniline

(2) 2, 4, 6-Tribromo aniline

(3) 4-Bromo aniline

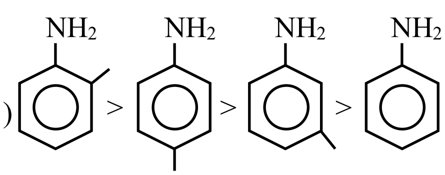
(4) 3-Bromo aniline

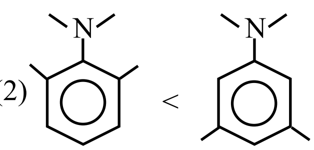
81. Match the compounds given in List-I with List-II and select the suitable option using the code given below:

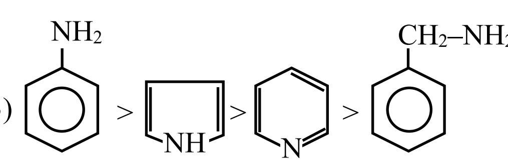
List-I			**List-II**
(1)	Benzaldehyde	**(i)**	Phenolphthalein
(2)	Phthalic anhydride	**(ii)**	Benzoin condensation
(3)	Phenyl benzoate	**(iii)**	Oil of wintergreen
(4)	Methyl salicylate	**(iv)**	Fries rearrangement

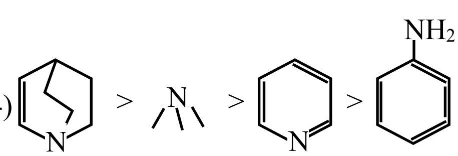
Code	**(1)**	**(2)**	**(3)**	**(4)**
(1)	(iv)	(i)	(iii)	(ii)
(2)	(iv)	(ii)	(iii)	(i)
(3)	(ii)	(iii)	(iv)	(i)
(4)	(ii)	(i)	(iv)	(iii)

82. Which is correct basicity order of following compounds :-

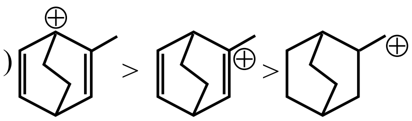
(1) 

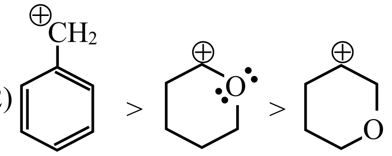
(2) 

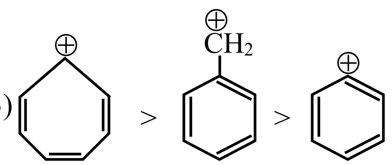
(3) 

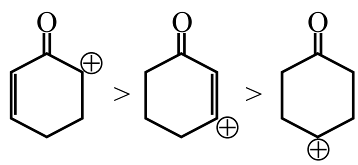
(4) 

83. Which is correct stability order of following carbocation :-

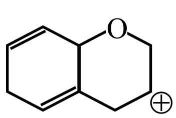
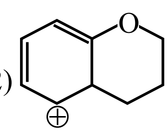
(1) 

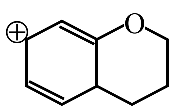
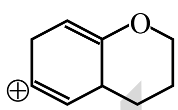
(2) 

(3) 

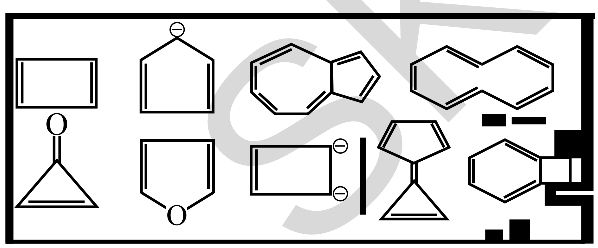
(4) 

84. Which of the following is most stable cation?

(1)  (2) 

(3)  (4) 

85. Number of aromatic compounds in given compounds :-



(1) 6 (2) 7 (3) 4 (4) 5

86. How many groups show +M effect

$-NO_2$, $-COOCH_3$, $-NH-COCH_3$, $-F$,

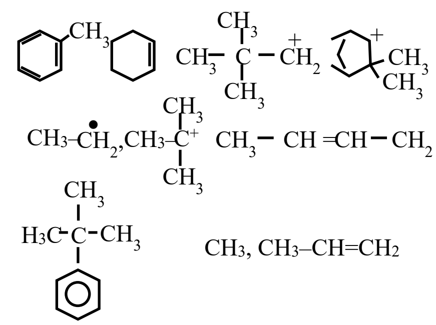
$-\overset{\ominus}{C}H_2$, $-CH_3$, $-CH(CH_3)_2$, $-C(=O)-O^{\ominus}$

$-CONH-CH_3$

(1) 5 (2) 6

(3) 4 (4) 7

87. How many compounds will show hyperconjugation effect



(1) 5 (2) 6 (3) 7 (4) 8

88. Correct order stability of resonating structure

(A) $CH_2=CH-CH=CH-\ddot{N}H-CH_3$

(B) $\overset{\ominus}{C}H_2=CH-CH=CH-\overset{\ominus}{N}H-CH_3$

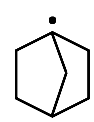
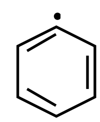
(C) $\overset{\ominus}{C}H_2=\overset{\oplus}{C}H-CH=CH-\ddot{N}H-CH_3$

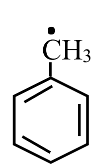
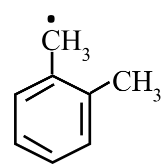
(D) $\overset{\oplus}{C}H_2=\overset{\ominus}{C}H-CH=CH-\ddot{N}H-CH_3$

(1) A > B > C > D (2) B > A > C > D

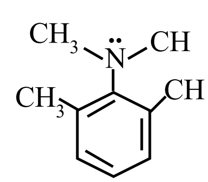
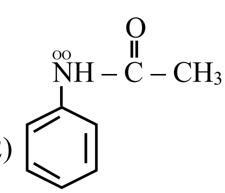
(3) A > B > D > C (4) A > C > D > B

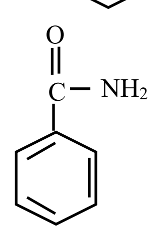
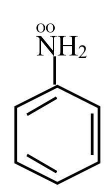
89. Which of the following is most stable carbon free radical

(1)  (2) 

(3)  (4) 

90. Which of the following benzene ring has highest electron density

(1)  (2) 

(3)  (4) 

BIOLOGY

91. Read statement regarding cardiac system and choose the correct option

A. Human heart is an ectodermal derivative.

B. Mitral valve lies at the left upper corner of the right atrium.

C. SAN is located on the right upper corner of the right atrium.

D. Certain factors released by the tissues at the site of injury also can initiate coagulation.

(1) A and B (2) A and D

(3) C and D (4) B and C

92. Properties of leucocytes are

A. They are nucleated

B. They are enucleated like RBC

C. They are 6000-8000 mm^{-3} of blood

D. They are long lived

E. They are short lived

(1) A, C and E (2) B, D and E

(3) A, D and E (4) A, C and D

93. Mark the correct option for the given statements:

A. Contraction of _____ increases the volume of the thoracic chamber in the antero-posterior axis

B. Contraction of _____ lifts up the ribs and the sternum and causes an increase in the volume of the thoracic chamber in the dorso- ventral axis

(1) A-Diaphragm, B-External intercostal muscles

(2) A-External intercostal muscles, B- Diaphragm

(3) A-Diaphragm, B-Internal intercostal muscles

(4) A-External intercostal muscles, B-Internal intercostal muscles

94. Match the following columns and choose the correct option:

Column-I (% of CO_2 transported)		**Column-II (Form of CO_2)**	
(A)	70	(i)	Dissolved in plasma
(B)	20-25	(ii)	Carbaminohaemoglobin
(C)	5-10	(iii)	Bicarbonates

(1) A-(i), B-(ii), C-(iii)

(2) A-(ii), B-(i), C-(iii)

(3) A-(iii), B-(ii), C-(i)

(4) A-(ii), B-(iii), C-(i)

95. What is true about RBCs in humans?

(A) RBCs are destroyed in the spleen

(B) A healthy adult man has 6000- 8000 mm^{-3} RBCs in blood

(C) In mammals, they are biconcave in shape.

(D) They have a red coloured, iron containing complex protein called haemoglobin

(1) A and B (2) A, B and D

(3) A, C and D (4) A, B, C and D

96. Which statement correctly suggests that most of the oxygen is transported from the lungs to the tissues, combined with the haemoglobin rather than dissolved in the blood plasma?

(1) Oxygen carrying capacity of the whole blood is much higher than that of plasma and oxygen content of the blood after leaving the lungs is greater than that of the blood entering the lungs.

(2) Haemoglobin can combine with oxygen.

(3) Oxyhaemoglobin can dissociate into haemoglobin and oxygen.

(4) Increase in the CO_2 concentration decrease the oxygen affinity of haemoglobin.

97. Read the following statements and choose the correct option:

Statement I- Spirometer cannot measure the volume of air remaining in the lungs after forced expiration

Statement II- Functional residual capacity includes ERV+RV

(1) Both statements are correct

(2) Both statements are incorrect

(3) Only statement I is correct

(4) Only statement II is correct

98. Choose the correct statements regarding muscle proteins

A. Actin is a thin filament and made up of two F-actins

B. The complex protein, tropomyosin is distributed at regular intervals of troponin

C. Myosin is a thick filament which is not a polymerized protein

D. The globular head of meromyosin consists of Light Meromyosin (LMM)

(1) A, B and C (2) A, B and D

(3) Only A (4) B and D

99. SAN generates an action potential which stimulates both the …A… to undergo a simultaneous contraction called …B…. This increases the flow of the blood into the ventricles by about …C… percent.

Choose the correct option for A, B and C

(1) A-atria, B-atrial systole, C-30

(2) A-ventricle, B-atrial systole, C-30

(3) A-atria, B-atrial systole, C-50

(4) A-atria, B-atrial diastole, C-50

100. Which of the following is the correct statement?

(1) Henle's loop is longer in juxta medullary nephrons

(2) Malpighian corpuscle, PCT and DCT of the nephron are situated in the cortical region of the kidney whereas the loop of Henle dips into the medulla

(3) Cortical nephrons are most abundant in our kidney

(4) All statements are correct

101. The osmolarity of interstitial fluid in kidneys is __A__ in the cortex and __B__ in the inner medulla:

	A	B
(1)	300 $mOsmolL^{-1}$	600 $mOsmolL^{-1}$
(2)	600 $mOsmolL^{-1}$	1000 $mOsmolL^{-1}$
(3)	300 $mOsmolL^{-1}$	1200 $mOsmolL^{-1}$
(4)	1200 $mOsmolL^{-1}$	300 $mOsmolL^{-1}$

102. Select the correct pathway for the passage of urine in humans

(1) Renal vein → Renal ureter → Bladder → Urethra
(2) Collecting tubule → Ureter → Bladder → Urethra
(3) Pelvis → Medulla → Bladder → Urethra
(4) Cortex → Medulla → Bladder → Ureter

103. Which of the following is an incorrect statement?

(1) The flow of filtrate in the two limbs of Henle's loop is in opposite directions and thus forms a counter current
(2) The flow of blood through the two limbs of vasa recta is also in a counter current pattern
(3) The proximity between Henle's loop and vasa recta, as well as the counter current in them, helps in maintaining decreasing osmolarity towards the inner medullary interstitium
(4) An increasing osmolarity gradient is maintained from the cortex to the medulla by NaCl and urea

104. Match the column

	Column I		Column II
(A)	PCT	(i)	Formation of concentrated urine
(B)	Tubular secretion	(ii)	Filtration of blood
(C)	Henle's loop	(iii)	Reabsorption of 70-80% of electrolytes
(D)	Counter-current mechanism	(iv)	Ionic balance
(E)	Renal corpuscle	(v)	High osmotic concentration gradient

(1) A-(iii), B-(v), C-(iii), D-(ii), E-(i)
(2) A-(iii), B-(iv), C-(i), D-(v), E-(ii)
(3) A-(i), B-(iii), C-(ii), D-(v), E-(iv)
(4) A-(iii), B-(i), C-(iv), D-(v), E-(ii)

105. Match the columns I and II and choose the correct option.

	Column I		Column II
A.	Cranial meninges	i.	Outer layer which is in contact with cranium
B	Dura mater	ii.	Inner layer which is in contact with brain tissue
C.	Arachnoid	iii.	Brain is covered by membrane inside the skull
D.	Pia mater	iv.	A very thin middle layer

(1) A-(ii), B-(i), C-(ii), D-(iii)
(2) A-(ii), B-(i), C-(iii), D-(iv)
(3) A-(i), B-(iii), C-(ii), D-(iii)
(4) A-(iii), B-(i), C-(iv), D-(ii)

106. Match the columns I and II and choose the correct

	Column I		Column II
A.	A fall in GFR can activate	i.	180 L/ per day
B.	Volume of filtrate formed	ii.	Renal tubules
C.	Amount of urine released	iii.	JG cells of kidney
D.	Tubular reabsorption	iv	1.5 L per day

(1) A-(iv), B-(ii), C-(iii), D-(i)
(2) A-(iv), B-(i), C-(ii), D-(iii)
(3) A-(iii), B-(i), C-(iv), D-(ii)
(4) A-(i), B-(ii), C-(iii), D-(iv)

107. A special sensitive region called juxtaglomerular apparatus (JGA) is formed by certain cells of:

A. Proximal convoluted tubule
B. Afferent arteriole
C. Distal convoluted tubule

(1) A and B only (2) B and C only
(3) A, B and C (4) A and C only

108. Among the given hormones which needs secondary messenger-

A. Glucagon B. Adrenaline
C. Steroid hormone D. Iodothyronine

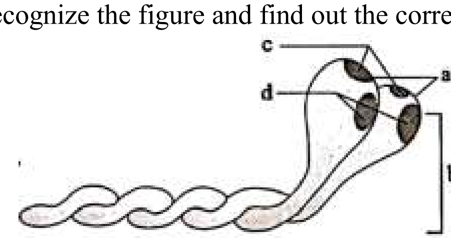
(1) A and C (2) C and D
(3) A and B (4) D and A

109. Match the following columns and choose the correct option:

	Column-I		Column-II
(A)	Vital capacity	(i)	ERV+TV
(B)	Functional residual capacity	(ii)	IRV+TV
(C)	Inspiratory capacity	(iii)	TV+IRV+ERV
(D)	Expiratory capacity	(iv)	ERV+RV

(1) A-(i), B-(iii), C-(ii), D-(iv)
(2) A-(iii), B-(iv), C-(ii), D-(i)
(3) A-(i), B-(iii), C-(iv), D-(ii)
(4) A-(ii), B-(i), C-(iii), D-(iv)

110. Recognize the figure and find out the correct matching.



(1) a - HMM, b - LMM, c - ATP Binding sites, d - Myosin binding sites
(2) a - Head, b - Cross bridge, c - Actin binding sites, d - ATP binding sites
(3) a - Head, b - Cross arm, c - ATP binding sites, d - actin binding sites
(4) a - Head, b - Cross arm, c - Actin binding sites, d - ATP Binding sites

111. Blood does not become acidic although it carries CO_2 because
(1) CO_2 is continuously diffused through tissues
(2) CO_2 combines with H_2O to form H_2CO_3
(3) Hemoglobin acts as buffer to resist change in pH of blood.
(4) CO_2 is absorbed by WBC

112. ____A____ is a tuft of capillaries formed by the ____B__ arteriole - a fine branch of __C__. Blood from ____A____ is carried away by an ____D____ arteriole. Identify A, B, C & D respectively.
(1) Glomerulus, afferent, renal vein and efferent
(2) Glomerulus, efferent, renal artery and afferent
(3) Glomerulus, afferent, renal artery and efferent
(4) Glomerulus, efferent, renal vein and afferent

113. Fibrous joints in humans
(A) Allows movement in any direction
(B) Allows little movement
(C) Don't allow any movement
(D) Is shown by the flat skull bones.
(1) A and D only (2) A and B only
(3) C and D only (4) B and D only

114. Prothrombin is
(1) Present in inactive state in plasma of blood
(2) A kind of hormone
(3) Changed to thrombin by Thrombokinase
(4) Both (1) and (3)

115. O_2 diffuses into …A… and forms HCO_3^- and H^+. At the …B… site where pCO_2 is low, the reaction proceeds in the opposite direction.
(1) A-WBC, B-diffusion
(2) A-RBC, B-alveolar
(3) A-RBC, B-tissue
(4) A- tissue cells, B-alveolar

116. Which of the following layers are present in the adrenal cortex from outer to Inner?
(1) Zona reticularis, zona fasciculata, zona glomerulosa
(2) Zona fasciculata, zona glomerulosa, zona reticularis
(3) Zona glomerulosa, zona reticularis, zona fasciculata
(4) Zona glomerulosa, zona fasciculata, zona reticularis

Direction-(Q.117 to Q.119)- Following questions consist of two statements – Assertion (A) and Reason (R). Answer these questions selecting the appropriate option given below:
(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true, but R is not the correct explanation of A
(C) A is true, but R is false
(D) A is false, but R is true

117. **Assertion (A):** Vasa recta is a minute vessel running parallel to the loop of Henle.
Reason (R): It remains in close proximity of loop of Henle to make counter current of blood and filtrate
(1) A (2) B (3) C (4) D

118. **Assertion (A):** Steroid hormones like cortisol and testosterone usually bind to intracellular receptors to exert their effects.
Reason (R): Being lipid-soluble, these hormones can easily diffuse across the cell membrane to enter the cytoplasm or nucleus.
(1) A (2) B (3) C (4) D

119. **Assertion (A):** Calcium is required for skeletal muscle contraction.
Reason (R): Calcium outflux causes release of acetylcholine at neuromuscular junction.
(1) A (2) B
(3) C (4) D

120. Match Column I with Column II:
Choose the correct answer from the options given below:

	Column I		Column II
A	Exophthalmic goitre	i	Diabetes Mellitus
B	Acromegaly	ii	Hypo secretion of thyroid hormone and stunted growth
C	Hyper -glycemia	iii	Hyper secretion of thyroid hormone & protruding eye balls.
D	Cretinism	iv	Excessive secretion of growth hormone

(1) A – (i), B – (iii), C – (ii), D – (iv)
(2) A – (iv), B – (ii), C – (i), D - (iii)
(3) A – (iii), B – (iv), C – (ii), D – (i)
(4) A – (iii), B – (iv), C – (i), D - (ii)

121. Joints are classified into three major types. They are
A. Fibrous joint B. Hinge joint
C. Cartilaginous joint D. Pivot joint
E. Synovial joint
(1) A, C and E (2) B, C and D
(3) A, B and C (4) C, D and E

122. **Statement I:** At a chemical synapse, the membranes of the pre- and post-synaptic neurons are separated by a fluid filled space called synaptic cleft.
Statement II: A nerve impulse is transmitted from one neuron to another through junctions called synapses.
(1) Statement I is correct but Statement II is incorrect.
(2) Both statements are correct.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statements are incorrect.

123. Match the following columns

Column-I		Column- II	
(A)	FSH	(i)	Synthesis of milk in breast
(B)	MSH	(ii)	Regulate the pigmentation
(C)	ADH	(iii)	Transported axonally to neurohypophysis
(D)	Prolactin	(iv)	Stimulate the development of ovarian follicle

(1) A-(i), B-(iii), C-(ii), D-(iv)
(2) A-(iv), B-(ii), C-(iii), D-(i)
(3) A-(i), B-(iii), C-(iv), D-(ii)
(4) A-(ii), B-(i), C-(iii), D-(iv)

124. Consider the following statements A, B, C and D and state from the options given whether they are True (T) or False (F):

A. Erythroblastosis foetalis can occur when an Rh^- mother is pregnant with an Rh^+ foetus.
B. Cardiac output of an athlete will be much lower than that of an ordinary man.
C. In an ECG, by counting the number of QRS complexes that occur in a given time period, one can determine the heart beat rate of an individual.
D. Parasympathetic nervous system decreases heart rate when stimulated.

	A	B	C	D
(1)	F	F	T	T
(2)	T	F	T	F
(3)	T	T	F	T
(4)	T	F	T	T

125. Arrange the given steps of respiration mechanism in the order, they occur in the human body

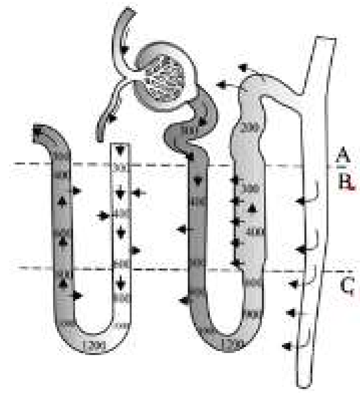
I. Breathing or pulmonary ventilation
II. Diffusion across the alveolar membrane
III. Transport of gases by blood
IV. Utilisation of O_2 by cells
V. Diffusion of O_2 and CO_2 between blood and tissues

(1) I → II → III → IV → V
(2) I → II → III → V → IV
(3) I → III → II → V → IV
(4) I → III → II → IV → V

126. Select the incorrect statement:

(1) Left atrium receives oxygenated blood from the lungs through two pairs of pulmonary arteries
(2) All four chambers of the heart are relaxed during joint diastole
(3) Volume of blood pumped out by each ventricle per minute is called stroke volume
(4) Both (1) and (3)

127. The given diagram showing counter current mechanism. Which part of kidney is involved in this mechanism (A-C)?

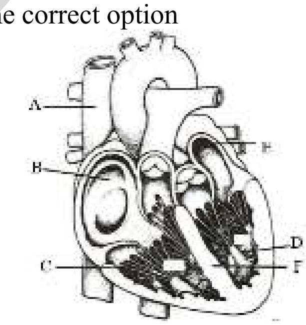


(1) A- Inner medulla, B- Outer medulla, C- Cortex
(2) A-Cortex, B- Outer medulla, C- Inner medulla
(3) A- Cortex, B- Inner medulla, C- Outer medulla
(4) A- Inner medulla, B- Cortex, C- Outer medulla

128. Bile juice is released by gali bladder, under the influence of an intestinal hormone called -

(1) Seretin (2) Cholecystolinin
(3) GIP (4) Gatrin

129. Identify A to F in the given diagram of human heart and choose the correct option



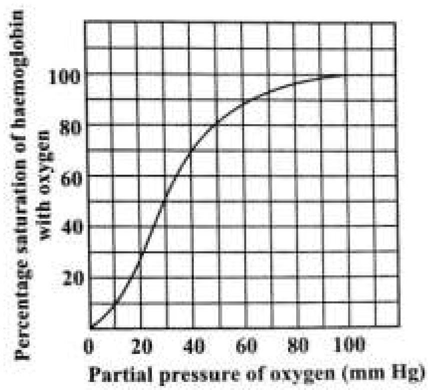
(1) A-Vena cava, B-Right atrium, C-Left atrium, D-Right ventricle, E-Left ventricle, F-Interventricular septum
(2) A-Vena cava, B-Right atrium, C-Right ventricle, D-Left ventricle, E-Left atrium, F- Interventricular septum
(3) A-Vena cava, B-Right atrium, C-Right ventricle, D-Left atrium, E-Left ventricle, F- Interventricular septum
(4) A-Vena cava, B-Left atrium, C-Right ventricle, D-Left ventricle, E-Right atrium, F-Interventricular septum

130. Which of the following is an incorrect statement?

(1) P-wave represents depolarisation of the atria
(2) Ventricular systole begins shortly after Q-wave
(3) T-wave represents the return of the ventricles from excited to normal state
(4) The end of T-wave marks the end of atrial systole.

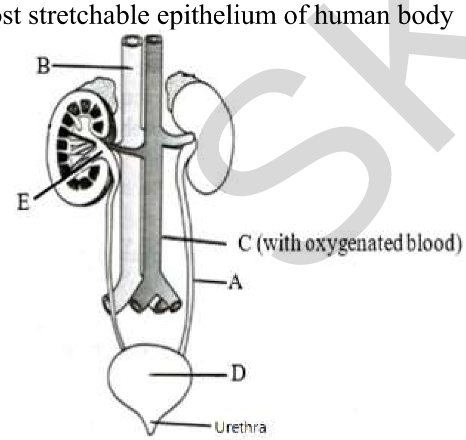
131. Read the given statements and choose the correct option-
Statement I- The SA node generates an action potential which stimulates both the ventricles to undergo simultaneous relaxation
Statement II- Ventricular diastole increases ventricular pressure, causing closure of the tricuspid and bicuspid valves
(1) Both statements are correct
(2) Both statements are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

132. Shifting of the curve to right takes place in the case of



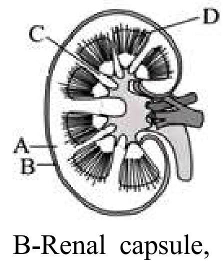
(1) Rise in pCO_2
(2) Fall in pH
(3) Rise in temperature
(4) All of the above

133. Which of the following structure (A to E) is lined by the most stretchable epithelium of human body



(1) A
(2) B
(3) D
(4) E

134. Given diagram is longitudinal section of kidney. Identify the labeled (A-D) parts?



(1) A-Cortex, B-Renal capsule, C-Renal column, D-Medullary pyramids
(2) A-Renal column, B-Medullary pyramids, C- Renal capsule, D- Cortex
(3) A-Medullary pyramids, B-Renal column, C-Cortex, D-Renal capsule
(4) A-Renal capsule, B-Cortex, C-Medullary pyramids, D-Renal column

135. Match the following columns

Column-I		Column- II	
(A)	Protein hormones	(i)	Epinephrine
(B)	Steroid hormones	(ii)	Testosterone, progesterone
(C)	Iodothyronines hormones	(iii)	Thyroxine
(D)	Amino acid derivatives hormones	(iv)	Insulin and glucagon

(1) A-(i), B-(iii), C-(ii), D-(iv)
(2) A-(iv), B-(iii), C-(i), D-(ii)
(3) A-(iv), B-(ii), C-(iii), D-(i)
(4) A-(ii), B-(i), C-(iii), D-(iv)

136. In meromyosin, the globular head:
A. Has the binding site for ATP
B. Has the binding site for actin
C. Is a component of light meromyosin
D. Is a component of heavy meromyosin
(1) A and B only
(2) B and C only
(3) A, B and D only
(4) A, B, C and D

137. Given below are four statements (i-iv) regarding human blood circulatory system.
(i) Arteries are thick-walled and have narrow lumen as compared to veins.
(ii) Angina is acute chest pain when the blood circulation to the brain is reduced.
(iii) Persons with blood group AB can donate blood to any persons with any blood group under ABO system.
(iv) Calcium ions play a very important role in blood clotting.
Which two of the above statements are correct?
(1) (i) and (iv)
(2) (i) and (ii)
(3) (ii) and (iii)
(4) (iii) and (iv)

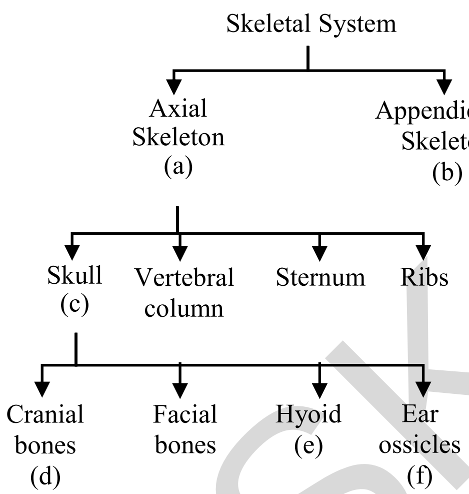
138. Which of the following statement is incorrect?
(1) A person of 'O' blood group has anti 'A' and anti 'B' antibodies in his blood plasma
(2) A person of 'B' blood group cannot donate blood to a person of 'A' blood group
(3) Blood group is designated on the basis of the presence of antibodies in the blood plasma
(4) A person of AB blood group is universal recipient

139. Identify the correct match from column-I and column-II and select the right option:

	Column-I		**Column-II**
(A)	Cerebrum	(i)	Biological Clock
(B)	Thalamus	(ii)	Will power
(C)	Hypothalamus	(iii)	Relay centre
(D)	Medulla oblongata	(iv)	Respiration

(1) A-(ii), B-(i), C-(iii), D-(iv)
(2) A-(iii), B-(ii), C-(iv), D-(i)
(3) A-(ii), B-(iii), C-(i), D-(iv)
(4) A-(i), B-(ii), C-(iv), D-(iii)

140. Recognise the figure and find out the number of bones in specified regions a, b, c, d, e and f.



(1) a-126, b-80, c-14, d-6, e-8, f-3
(2) a-80, b-126, c-29, d-14, e-1, f-3
(3) a-80, b-126, c-29, d-8, e-1, f-6
(4) a-126, b-80, c-29, d-14, e-2, f-6

141. Glucose and amino acids in the filtrate are reabsorbed by tubular epithelial cells through -
(1) Active transport
(2) Passive transport
(3) Osmosis
(4) Both (2) and (3)

142. During contraction of skeletal muscles:
(1) Size of I- band decreases, while A-band remains the same
(2) Size of both I-band and A-band increases
(3) Size of I-band remains the same while that of A-band decreases
(4) Size of both I-band and A-band decreases

143. Fill in the blanks with correct options:
(A) Each actin (thin) filament is made of __I__ 'F' (filamentous) actins helically wound to each other
(B) Two filaments of another protein __II__ also run close to 'F' actins throughout its length.
(C) In the resting state, a subunit of __III__ masks the active binding sites for myosin on actin filaments
(D) Many monomeric proteins called __IV__ constitute one thick filament
(1) I- Four, II-Troponin, III-Tropomyosin, IV- Actin
(2) I- Two, II- Tropomyosin, III- Troponin, IV- Meromyosins
(3) I- Three, II- Tropomyosin, III- Troponin, IV- Actin
(4) I- Two, II- Troponin, III- Tropomyosin, IV- Meromyosin

144. Find out the correct sequence of events that take place during muscle contraction:
A. Hydrolysis of ATP
B. Formation of cross-bridge
C. Release of calcium ions from sarcoplasmic reticulum.
D. Release of neurotransmitter at neuromuscular junction
E. Generation of action potential in the sarcolemma
F. Binding of calcium with subunit of troponin
(1) A→ C→ D→ F→ E→ B
(2) C→ D→ E→ A→ B→ F
(3) D→ E→ C→ F→ A→ B
(4) D→ E→ C→ A→ F→ B

145. **Statement I-** The dorsal, flat, triangular body of the scapula has a slightly elevated ridge called the spine which projects as a flat, expanded process called the acromion.
Statement II- Below the acromion is a depression called the acetabulum, which articulates with the head of the humerus to form the shoulder joint
(1) Both statements are correct
(2) Both statements are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

146. **Statement I:** Neutrophils, eosinophils and basophils are different types of granulocytes, while lymphocytes and monocytes are the agranulocytes.
Statement II: Basophils are the most abundant cells (60- 65%) of the total WBCs and neutrophils are the least (0.5-1%) among them.
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both statements are correct.
(4) Both Statements are incorrect.

147. The thymus gland is a lobular structure located on the …A… side of the …B… and aorta. The thymus plays a significant role in the development of …C… system
Choose the correct combination of A, B and C
(1) A-ventral, B-heart, C-immune
(2) A-lateral, B-kidney, C-circulatory
(3) A-dorsal, B-heart, C-immune
(4) A-dorsal, B-parathyroid, C-circulatory

148. False ribs are ventrally attached to
(1) Sternum through hyaline cartilage
(2) Seventh rib through hyaline cartilage
(3) Vertebral column
(4) Sternum through fibrous cartilage

149. During ventricular systole
(1) Ventricle receives blood from respective auricles
(2) Auricle pumps blood into ventricles
(3) Auriculo-ventricular valves remains open.
(4) Oxygenated blood is pumped into the aorta and deoxygenated blood is pumped into the pulmonary artery

150. Which of the following is not the feature of red muscle fibres -
(1) They have plenty of mitochondria
(2) They have high content of Myoglobin
(3) They have high amount of Sarcoplasmic reticulum
(4) They are called aerobic muscles

151. **Statement I-** Force generated by smooth muscles is used to carry out movement through joints
Statement II- Joints act as fulcrum during movement
(1) Both statements are correct
(2) Both statements are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

152. Striations in the striated muscles are due to-
(1) Absence of myofilaments
(2) Presence of myofilaments
(3) Specialized arrangement of myofilaments
(4) Projections of myosin

153. Match the following and mark the correct option:

	Column-I		Column-II
(A)	Sternum	(i)	Ball and socket joint
(B)	Glenoid cavity	(ii)	Between vertebrae
(C)	Freely movable joint	(iii)	Pectoral girdle
(D)	Cartilaginous joint	(iv)	Flat bones

(1) A-(ii), B-(i), C-(iii), D-(iv)
(2) A-(iv), B-(iii), C-(i), D-(ii)
(3) A-(ii), B-(i), C-(iv), D-(iii)
(4) A-(iv), B-(i), C-(ii), D-(iii)

154. **A.** Muscular growth, aggressiveness, low pitch voice.
B. Anabolic effect on protein and carbohydrate metabolism.
C. Stimulates male gamete formation
These functions refer to?
(1) Oestrogens (2) Progesterone
(3) Testosterone (4) Adrenal androgens

155. Which of the statements about the mechanism of muscle contraction are correct?
(A) During muscle contraction, I-band remains unchanged.
(B) Repeated activation of the muscles can lead to lactic acid accumulation
(C) Acetycholine is released when the neural signal reaches the motor end plate
(D) Muscle contraction is initiated by a signal sent by the CNS via a sensory neuron
(1) A and B (2) C and D
(3) A and C (4) B and C

156. **Assertion:** Open circulatory system is found in most arthropods.
Reason: Arthropods contain haemolymph which directly bathes internal tissues and organs.
(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
(3) Assertion is correct but Reason is incorrect. (4) Both Assertion and Reason are incorrect.

157. Match the following columns.

Column I (Animals)		Column II (Types of Breathing)	
A.	Frog	(i)	Network of trachae
B.	Fishes	(ii)	Pulmonary respiration
C.	Insects	(iii)	Branchial respiration
D.	Birds and mammals	(iv)	Cutaneous respiration

	A	B	C	D
(1)	(i)	(ii)	(iii)	(iv)
(2)	(iv)	(iii)	(i)	(ii)
(3)	(iv)	(iii)	(ii)	(i)
(4)	(iv)	(ii)	(iii)	(i)

158. The Na^+ - K^+ pump in the axonal membrane:
(1) Is operative during the resting stage
(2) Transports $3Na^+$ outwards and $2K^+$ into the cell
(3) Is operative during repolarisation and trans-ports $3Na^+$ and $2K^+$ into the cell
(4) Both (1) and (2) correct.

159. Glomerular filtrate is:-
(1) Blood minus blood corpuscles and plasma protein
(2) Isotonic to blood plasma
(3) Contains essential as well as waste products.
(4) All of the above

160. Which of the following statements are correct for chemical synapses?
A. Pre and post-synaptic neurons are separated by fluid filled synaptic cleft
B. The membranes of the pre-synaptic and post-synaptic neurons are in close proximity
C. No neurotransmitters is involved.
D. Shows slower conduction.
(1) A and B (2) B and C
(3) A and D (4) B and D

161. Which of the following are the secondary messengers?
A. Cyclic AMP B. IP_3 C. Ca^{2+}
(1) A and B (2) B and C
(3) A and C (4) A, B and C

162. Which of the following is the correct option with respect to the function of the following gonadotropins?

	FSH	LH
(1)	Release of ovum from Graafian follicle	Development of follicle in menstrual cycle
(2)	Release of progesterone from corpus luteum	Release of oestrogen from developing follicle
(3)	Release of androgens from Leydig cells	Promotes spermatogenesis
(4)	Stimulates growth of ovarian follicles	Induces ovulation of mature follicles

163. SA node is called the pacemaker of heart because
(1) It can change the contractile activity generated by AV node
(2) It delays the transmission of impulse between the atria and ventricles
(3) It gets stimulated when it receives neural signals
(4) It initiates and maintains the rhythmic contractile activity of heart

164. Mark the incorrect statements
(1) Cigarette smoking may lead to inflammation of bronchi
(2) Pons consists of tracts of fibres that interconnect different regions of the brain.
(3) Workers in grinding and stone breaking industries may suffer from lung fibrosis
(4) About 70% of CO_2 is carried out by haemoglobin as carbominohaemoglobin

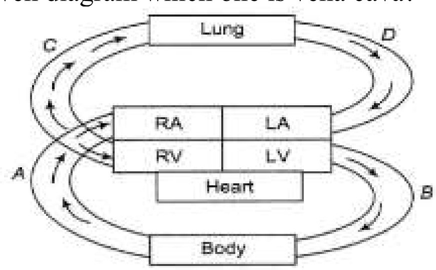
165. Consider the following statements and Select the option with correct statement.
A. Pituitary gland is attached and works under regulation of hypothalamus.
B. In humans, pars intermedia is almost merged with pars distalis.
C. Oxytocin is synthesised by hypothalamus and is transported axonally to neurohypophysis.
(1) Only A (2) A and B
(3) A, B, and C (4) A and C

166. During muscle contraction, ATP provides energy for
(1) Cross bridge detachment
(2) Building up action potential
(3) Releasing Ca^{2+} from sarcoplasmic reticulum
(4) All of these

167. The hyposecretion of which hormone leads to loss of sodium and water through urine, low blood pressure (hypotension)?
(1) Cortisol
(2) Aldosterone
(3) Hormones of adrenal medulla
(4) ANF

168. Choose the statements which correctly indicates the functioning of thyroid hormones.
A. Regulation of the basal metabolic rate.
B. Stimulate the process of RBCs formation.
C. Regulating the sleep wake cycle.
D. Maintenance of pH and ionic balance.
(1) A, B and D (2) A and B
(3) A, B, C and D (4) C and D

169. In given diagram which one is vena cava?



(1) A (2) B (3) C (4) D

Direction-(Q. 170 to Q. 172)
(A) If both A and R are true and R is the correct explanation of A
(B) If both A and R are true, but R is not the correct explanation of A
(C) If A is true, but R is false
(D) If A is false, but R is true

170. **Assertion:** JGA gets activated in a person suffering from loose motions.
Reason: Water and electrolytes intake is required in such a patient.
(1) A (2) B (3) C (4) D

171. **Assertion (A):** Diabetes insipidus is caused by deficiency of ADH.
Reason(R): ADH is secreted by posterior lobe of pituitary
(1) A (2) B (3) C (4) D

172. **Assertion(A):** In old persons, there is gradually weakening of immune system.
Reason(R): It is because of degeneration of thymus gland.
(1) A (2) B (3) C (4) D

173. When carbon dioxide concentration in blood increases, breathing becomes
(1) Shallower and slow
(2) There is no effect on breathing
(3) Slow and deep
(4) Faster and deeper

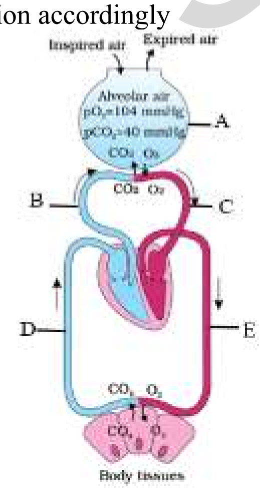
174. Match the following columns

Column-I		Column- II	
(A)	Ball and socket joint	(i)	Between humerus and pectoral girdle
(B)	Hinge joint	(ii)	Knee joint
(C)	Pivot joint	(iii)	Between atlas and axis
(D)	Gliding joint	(iv)	Between the carpals

(1) A-(i), B-(ii), C-(iii), D-(iv)
(2)A-(iv), B-(iii), C-(i), D-(ii)
(3) A-(ii), B-(i), C-(iv), D-(iii)
(4) A-(iv), B-(i), C-(ii), D-(iii)

175. Consider the following statements
A. Parathormone regulates the metabolism of calcium.
B. Oxytocin stimulates contraction of uterine muscles during child birth.
C. Grave's disease is caused by malfunctioning of adrenal gland.
D. Cortisol suppresses immune response in our body.
(1) A and C are true; B and D are false
(2) A and B are true; C and D are false
(3) A, B and D are true, only C false
(4) A, B and C are true; D only false

176. Identify A to E in the given diagram and choose the correct option accordingly



(1) A-Alveolus, B-Pulmonary artery,
C- Pulmonary vein, D-Systemic vein,
E- Systemic arteries
(2) A-Alveolus, B-Pulmonary vein,
C- Pulmonary artery, D-Systemic vein,
E- Systemic arteries
(3) A-Alveolus, B-Pulmonary vein,
C- Pulmonary artery, D-Systemic arteries,
E- Systemic vein
(4) A-Alveolus, B-Pulmonary vein,
C- Pulmonary artery, D-Systemic arteries,
E- Portal vein

177. Urine formed by kidneys is ultimately carried to …A… where it is stored till a voluntary signal is given by the …B…. This signal is initiated by …C… of urinary bladder as it gets filled with urine.
(1) A-urethra, B-CNS, C-PNS
(2) A-urinary bladder, B-CNS, C-stretching
(3) A-urethra, B-CNS, C-stretching
(4) A-urethra, B-CNS, C-ANS

178. Which of the following are the properties of cardiac muscles? except one.
(1) They never get fatigued
(2) They are non-striated
(3) They are involuntary in their functions
(4) They have intercalated discs

179. Mark the wrongly matched pair

	Source gland	Hormone	Deficiency disorder
(1)	Neurohypophysis	Vasopressin	Diabetes insipidus
(2)	Adrenal cortex	Corticosteroids	Addison's disease
(3)	Parathyroid glands	Parathormone	Tetany
(4)	Thyroid glands	Calcitonin	Acromegaly

180. Erythroblastosis fetalis can be prevented by administering anti-Rh antibodies to:
(1) Rh^- mother after delivery of Rh^+ baby
(2 Rh^+ mother after delivery of Rh^+ baby
(3) Rh^+ mother after delivery of Rh^- baby
(4) Both (2) and (3)

Answer Key

and

Solution

ANSWER KEY & SOLUTION

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	3	3	4	2	1	2	4	4	3	1	1	1	4	3
Que.	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
Ans.	2	1	4	4	4	2	1	2	3	3	3	4	2	2	1
Que.	**31**	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**	**41**	**42**	**43**	**44**	**45**
Ans.	2	1	4	3	2	4	2	2	1	1	1	3	4	4	3
Que.	**46**	**47**	**48**	**49**	**50**	**51**	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
Ans.	4	1	3	1	2	1	1	2	4	3	1	3	3	2	4
Que.	**61**	**62**	**63**	**64**	**65**	**66**	**67**	**68**	**69**	**70**	**71**	**72**	**73**	**74**	**75**
Ans.	4	2	2	3	2	2	4	3	1	3	2	1	1	3	2
Que.	**76**	**77**	**78**	**79**	**80**	**81**	**82**	**83**	**84**	**85**	**86**	**87**	**88**	**89**	**90**
Ans.	2	3	2	1	2	3	3	3	2	1	1	1	1	4	4
Que.	**91**	**92**	**93**	**94**	**95**	**96**	**97**	**98**	**99**	**100**	**101**	**102**	**103**	**104**	**105**
Ans.	3	1	1	3	3	1	1	3	1	4	3	2	3	2	4
Que.	**106**	**107**	**108**	**109**	**110**	**111**	**112**	**113**	**114**	**115**	**116**	**117**	**118**	**119**	**120**
Ans.	3	2	3	2	4	3	3	3	4	2	4	1	1	3	4
Que.	**121**	**122**	**123**	**124**	**125**	**126**	**127**	**128**	**129**	**130**	**131**	**132**	**133**	**134**	**135**
Ans.	1	2	2	4	2	4	2	2	2	4	2	4	3	1	3
Que.	**136**	**137**	**138**	**139**	**140**	**141**	**142**	**143**	**144**	**145**	**146**	**147**	**148**	**149**	**150**
Ans.	3	1	3	3	3	1	1	2	3	3	1	1	2	4	3
Que.	**151**	**152**	**153**	**154**	**155**	**156**	**157**	**158**	**159**	**160**	**161**	**162**	**163**	**164**	**165**
Ans.	4	3	2	3	4	1	2	4	4	3	4	4	4	4	3
Que.	**166**	**167**	**168**	**169**	**170**	**171**	**172**	**173**	**174**	**175**	**176**	**177**	**178**	**179**	**180**
Ans.	1	2	2	1	2	2	1	4	1	3	1	2	2	4	1

PHYSICS

1. (3)

Sol. As the particle starts its motion from the equilibrium position, so

$x = A \sin \omega t = A \sin \frac{2\pi}{T} t$; At $t = \frac{T}{12}$,

$x = A \sin \frac{2\pi}{T} \times \frac{T}{12} = \frac{A}{2}$

$\therefore$Kinetic energy,

$K = \frac{1}{2} m\omega^2 \left(A^2 - x^2\right) = \frac{1}{2} m\omega^2 \left(A^2 - \frac{A^2}{4}\right)$

$= \frac{3}{4}\left(\frac{1}{2} m\omega^2 A^2\right)$

and potential energy,

$U = \frac{1}{2} m\omega^2 x^2 = \frac{1}{2} m\omega^2 \left(\frac{A}{2}\right)^2$

$= \frac{1}{4}\left(\frac{1}{2} m\omega^2 A^2\right)$

Thus. $\frac{K}{U} = \frac{3}{1}$

2. (3)

Sol $4v^2 = 25 - x^2$

Differentiating it both sides, we get

$4\left(2v\frac{dv}{dt}\right) = -2x\frac{dx}{dt}$ or $4\frac{dv}{dt} = -x$

$\left(\because \quad \right)$

or $4a = -x$ or $a = -\frac{1}{4}x$ $\left(\because \quad \right)$

Comparing it with $a = -\omega^2 x$, we get $\omega = \frac{1}{4}$ or

$\omega = \frac{1}{2}$

$\therefore$ Time period, $T = \frac{2\pi}{\omega} = \frac{2\pi}{1/2} = 4\pi$

3. (3)

Sol. Kinetic energy when the displacement of the particle is x is given by $K = \frac{1}{2} m\omega^2 \left(A^2 - x^2\right)$

Potential energy at this instant is given by

$U = \frac{1}{2} m\omega^2 x^2$

As $K = \frac{1}{8} U \therefore \frac{1}{2} m\omega^2 \left(A^2 - x^2\right) = \frac{1}{8}\left(\frac{1}{2} m\omega^2 x^2\right)$

or $\left(A^2 - x^2\right) = \frac{x^2}{8}$ or $x = \frac{2\sqrt{2}}{3} A$

4. (4)

Sol. $\Delta\phi = (\omega_1 - \omega_2)t = \left(\frac{2\pi}{T} - \frac{2\pi}{5T/4}\right)T = \frac{2\pi}{5}$

5. (2)

Sol. Maximum velocity, $v_m = A\omega = A\left(\frac{2\pi}{T}\right)$

$\therefore T = \frac{2\pi A}{v_m} = 2 \times \frac{22}{7} \times \frac{7 \times 10^{-3}}{4.4} = 10^{-2} s = 0.01 s$

6. (1) Since

$y_1 = 10 \sin\left(10\pi t + \frac{\pi}{3}\right) m$

and $y_2 = 10 \sin\left(8\pi t + \frac{\pi}{4} + \frac{\pi}{2}\right) m$

Phase difference between the two particles is given as

$\Delta\phi = \left(10\pi t + \frac{\pi}{3}\right) - \left(8\pi t + \frac{3\pi}{4}\right)$

$\Delta\phi = 2\pi t - \frac{5\pi}{12}$ as given that t = 0.5 sec

Therefore, $\Delta\phi = \frac{7\pi}{12}$

7. (2)

Sol $x_1 = 10 \sin\left(3\pi t + \frac{\pi}{4}\right)$

and $x_2 = 5\left(\sin 3\pi t + \sqrt{3} \cos 3\pi t\right)$

$= 10\left(\frac{1}{2}\sin 3\pi t + \frac{\sqrt{3}}{2}\cos 3\pi t\right)$

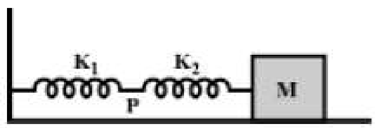
$x_2 = 10\left(\cos\frac{\pi}{3}\sin 3\pi t + \sin\frac{\pi}{3}\cos 3\pi t\right)$

$= 10 \sin\left(3\pi t + \frac{\pi}{3}\right)$

So, Amplitude $= \frac{10}{10} = 1:1$

8. (4)

Sol.



Here the two springs are in series. Let the mass M be pulled through small distance A towards right hand side. Due to it, let x_1 and x_2 be the extension in two springs.

When two springs are in series. the tension or the force is same for both the springs. So $k_1x_1 = k_2x_2$

or $x_2 = \frac{k_1x_1}{k_2}$

Total extension,

$$A = x_1 + x_2 = x_1 + \frac{k_1x_1}{k_2}; A = x_1\left(\frac{k_1+k_2}{k_2}\right)$$

$\therefore$Amplitude of point P = extension of spring

$$x_1 = \frac{k_2A}{k_1+k_2}$$

9. (4)

Sol. Since maximum velocity is $A\omega$ where ω is angular frequency.

$$\therefore A_1\omega_1 = A_2\omega_2 \text{ or } \frac{A_1}{A_2} = \frac{\omega_2}{\omega_1}$$

But $\omega = \sqrt{\frac{k}{m}} \therefore \frac{A_1}{A_2} = \sqrt{\frac{k_2}{k_1}}$

10. (3)

Sol. Here, density of bob, $\rho = \frac{4}{3}\times 1000 kgm^{-3}$ and density of water, $\sigma = 1000 kgm^{-3}$

In air, $T_0 = 2\pi\sqrt{\frac{1}{g}}$

In water, $T = 2\pi\sqrt{\frac{1}{g\left(1-\frac{\sigma}{\rho}\right)}} = 2\pi\sqrt{\frac{1}{g\left(1-\frac{3}{4}\right)}}$

$$= 2\left(2\pi\sqrt{\frac{1}{g}}\right) = 2T_0$$

11. (1)

Sol Here

$$y(x,t) = A\sin(kx-\omega t) \therefore \frac{\partial y}{\partial x} = kA\cos(kx-\omega t)$$

Now use the expression $\Delta P = -B\frac{\partial y}{\partial x}$ where B is the bulk modulus. This gives an expression for the pressure change

$$\Delta P = -BAk\cos(kx-\omega t)$$

Pressure amplitude. P_0= BAk

Also $\Delta P = -BAk\sin\left[(kx-\omega t)-\frac{\pi}{2}\right]$

also indicates a phase lag of $\frac{\pi}{2}$ with respect to displacement.

12. (1)

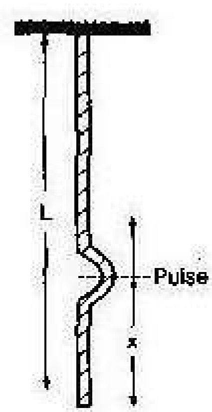
Sol Velocity of transverse wave along string is $v = \sqrt{\frac{T}{\mu}}$ where μ is the mass per unit length of string.

Here, $T = 500N, \mu = 0.2 kg/m$

$$\therefore v = \sqrt{\frac{500}{0.2}} = \sqrt{\frac{500\times 10}{2}} = \sqrt{2500} = 50 m/s$$

13. (1)

Sol. As the rope has mass and it is suspended vertically, tension in it will be different at different points. For a point at a distance x from the free end, tension will be due to the weight of the rope below it. If M is the mass of rope of length L, the mass of length x of the rope will be (M/L)x



$$\therefore T = \left[\frac{M}{L}x\right]g$$

and $v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{Mgx}{L(M/L)}} = \sqrt{gx}$

The tension and the velocity of the wave of the wave is different at different points. If at point x the wave travels a distance dx in time dt.

$$v = \frac{dx}{dt} \text{ or } \sqrt{gx} = \frac{dx}{dt}$$

$$\text{or} \int dt = \int \frac{dx}{\sqrt{gx}} \text{ or } t = \frac{1}{\sqrt{g}} \int_0^L x^{-1/2} dx$$

$$\text{or } t = 2\sqrt{\left(\frac{L}{g}\right)}$$

$$\text{As } L = 2.45 m \therefore t = 2\sqrt{\frac{2.45}{9.8}} = 1 s$$

14. (4)

Sol Here, $A_1 = A, A_2 = A, \phi = 120^0$

The amplitude of the resultant wave is

$$A_R = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\phi}$$

$$= \sqrt{A^2 + A^2 + 2AA\cos 120^0}$$

$$= \sqrt{A^2 + A^2 - A^2} = A \quad \left(\because \qquad = -\frac{1}{2}\right)$$

15. (3)

16. (2)

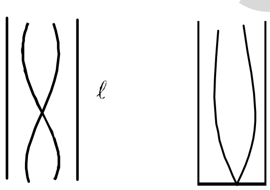
Sol. According to superposition principle,

$$y = y_1 + y_2 = a\sin(\omega t + kx) - a\sin(\omega t - kx)$$

$$y = 2a\sin kx \cos\omega t$$

17. (1)

Sol.



$\eta_1 = \frac{V}{2\ell}$ $\qquad$ $\eta_2 = \frac{V}{4\ell}$

no. of beat heard $n_1 - n_2 = \frac{V}{4\ell} = 4$

if length pipes are doubled. no of beats heard $n_1^1 - n_2^1 = \frac{V}{8\ell} = \frac{4}{2} = 2$

18. (4)

Sol $I = \frac{1}{2}\rho A^2 \omega^2 V$ put values

19. (4)

Sol Fundamental frequency, $\upsilon = \frac{1}{2\ell} \sqrt{T}$

$$\therefore \frac{\upsilon_2}{\upsilon_1} = \frac{\ell}{\ell} \qquad \frac{\ell}{\left(\ell \qquad \ell\right)} \sqrt{\left(\frac{\frac{44}{100}T_1}{T_1}\right)}$$

$$\text{or } \therefore \frac{\upsilon_2}{\upsilon_1} = \frac{100}{60} \times \frac{12}{10} = \frac{2}{1} \text{ or } \upsilon_2 : \upsilon_1 = 2:1$$

20. (4)

Sol. In spring mass system time period depends only on k and m.

21. (2)

Sol. The fundamental frequency of a string is

$$\upsilon = \frac{1}{2L}\sqrt{\frac{T}{\pi r^2 \rho}} = \frac{1}{2Lr}\sqrt{\frac{T}{\pi\rho}}$$

where the symbols have their values, we get

$$\therefore \frac{\upsilon_A}{\upsilon_B} = \left(\frac{L_B}{L_A}\right)\left(\frac{r_B}{r_A}\right)\left(\frac{T_A\rho_B}{T_B\rho_A}\right)^{1/2}$$

Substituting the given values, we get

$$\frac{\upsilon_A}{\upsilon_B} = \left(\frac{2L}{L}\right)\left(\frac{2r}{r}\right)\left(\frac{T}{2T}\frac{2\rho}{\rho}\right)^{1/2} = 4 \text{ or}$$

$$\upsilon_A = 4\upsilon_B$$

22. (1)

Sol As shown in diagram, this is 'sine' wave, then it should be A sin kx. Also at t = 0, all particle are at extreme positions, so it should be cos ωt

23. (2)

Sol.

$$R_A = \frac{V}{V_A}, R_B = \frac{V}{V_B}$$

as $V_A > V_B$, $R_A < R_B$

24. **(3)**

Sol. $F = -\frac{dU}{dx} = -8\sin 2x$

$a = \frac{F}{m} = -8\sin 2x\ (\because m = 1\text{ kg})$

For small oscillations, $\sin 2x \approx 2x$

i.e., $a = -16x$

Since, $a \propto -x$ the oscillations are SHM in nature.

$\therefore T = 2\pi\sqrt{\left|\frac{m}{k}\right|} = 2\pi\sqrt{\frac{1}{16}} = \frac{\pi}{2}$ sec.

25. **(3)**

Sol. $T = 2\pi\sqrt{\frac{l}{g}}; T' = 2\pi\sqrt{\frac{l}{g'}}$

$mg' = mg - V\rho_l g$

$= mg - \frac{m}{\rho_b}\rho_l g$

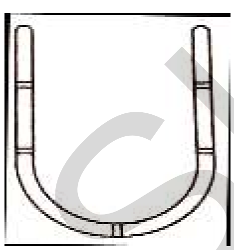
$= mg\left(1 - \frac{\rho_l}{\rho_b}\right)$

$g' = g(1-\rho)\ \left(\because \frac{\rho_l}{\rho_b} = \rho\right)$

$\Rightarrow T = 2\pi\sqrt{\frac{l}{g(1-\rho)}} = \frac{T}{\sqrt{1-\rho}}$

26. **(3)**

Sol. Consider the diagram in which a liquid column oscillates. In this case, restoring force acts on the liquid due to gravity. Acceleration of the liquid column can be calculated in terms of restoring force.



Restoring force, f = Weight of liquid column of height 2y

$\Rightarrow f = -(A \times 2y \times \rho) \times g = -2A\rho gy\ (\because m = \rho v)$

$\Rightarrow f \propto -y \Rightarrow$ Motion is SHM with force constant,

$k = 2A\rho g$.

$\Rightarrow$ Time period,

$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{A \times 2h \times \rho}{2A\rho g}} = 2\pi\sqrt{\frac{h}{g}}$

$T = 2\pi\sqrt{\frac{l}{g}}$, where $l = h$

which is independent of the density of the liquid.

27. **(4)**

Sol. $v = \omega\sqrt{A^2 - x^2}$ (In SHM)

or $v^2 = \omega^2A^2 - \omega^2x^2$

Comparing the given equation with the above equation, we get

$\omega^2 = 9$

$\therefore \omega = 3 = \frac{2\pi}{T}$

$\therefore T = \frac{2\pi}{3}$ units

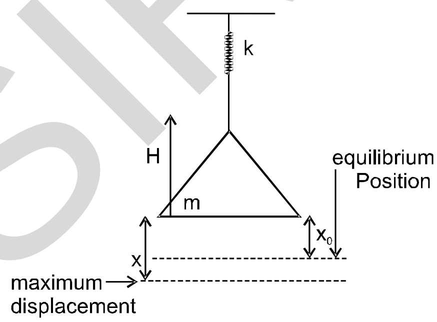
Also, $\omega^2A^2 = 144$

$\therefore A = 4$ units, $|a| = \omega^2x = (9)(3) = 27$ units

Displacement $\leq$ Distance

28. **(2)**

Sol.



velocity before collision = $\sqrt{2gH}$

pan is massless so velocity after collision

= $\sqrt{2gH}$ by energy conservation

$mg(x) + \frac{1}{2}m\left(\sqrt{2gH}\right)^2 = \frac{1}{2}Kx^2$

$Kx^2 - 2mgx - 2mgH = 0$

$x = \frac{mg}{K} + \frac{mg}{K}\sqrt{1 + \frac{2HK}{mg}}$

at equilibrium $Kx_0 = mg \Rightarrow x_0 = mg/K$

mean Amplitude = $x - x_0 = \frac{mg}{K}\sqrt{1 + \frac{2HK}{mg}}$

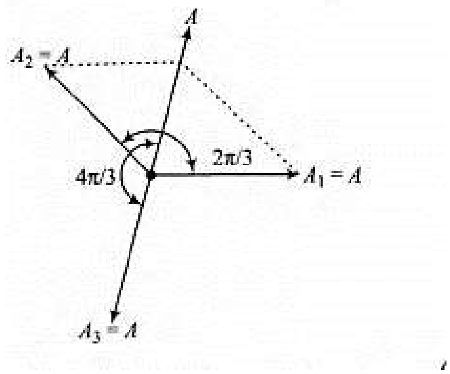
29. **(2)**

Sol. It we substitute, $x_1 + x_2 + x_3 = 0$

or $A\sin\omega t + A\sin\left(\omega t + \frac{2\pi}{3}\right) + B\sin(\omega t + \Phi) = 0$

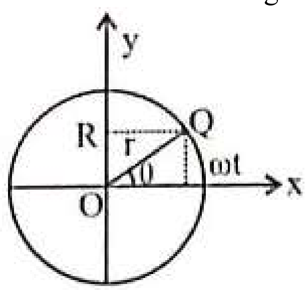
Then by applying simple mathematics we can prove that

$B = A$ and $\Phi = \frac{4\pi}{3}$



30. **(1)**

Sol. Let angular velocity of the particle executing circular motion is ω and when it is at Q makes an angle θ as shown in the diagram.



Clearly, $\theta = \omega t$

Now, we can write $OR = OQ\cos(90 - \theta)$

$= OQ\sin\theta = OQ\sin\omega t = r\sin\omega t$ [$\because OQ = r$]

$\Rightarrow\ x = r\sin\omega t = B\sin\omega t$

[$\because r = B$]

$x = B\sin\frac{2\pi}{T}t = B\sin\left(\frac{2\pi}{30}t\right)$

Clearly, this equation represents S.H.M.

31. **(2)**

Sol. If a pendulum is suspended in a lift and lift is moving downward with some acceleration a, then time period of pendulum is given by, $T = 2\pi\sqrt{\frac{l}{g-a}}$

In the case of free fall, a = g, then $T = \infty$ i.e., the time period of pendulum becomes infinite.

32. **(1)**

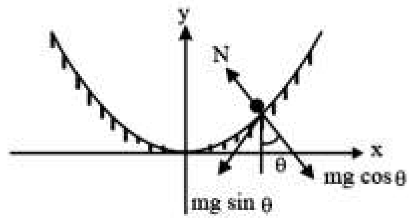
Sol. Potential energy is minimum (in this case zero) at mean position ($x \propto 0$) and maximum at extreme positions ($x = \pm A$)

At time t = 0, x = A. Hence P.E. should be maximum. There fore graph I is correct.

Further in graph III, P.E. is minimum at x = 0. Hence this is also correct.

33. **(4)**

Sol.



ma=-mg cos θ

$a = -g\sin\theta$ or $a = -g\tan\theta$...(i)

(as θ is small)

Now, $x^2 = 4ay$

$\therefore \frac{dy}{dx} = \frac{x}{2a}$

$\therefore a = -g\frac{x}{2a}$

$-\omega^2 x = -\frac{gx}{2a}$

$\omega = \sqrt{\frac{g}{2a}}$

34. **(3)**

Sol. At mean position → x = 0

So, $V = \omega\sqrt{A^2 - x^2} = \omega A$

And, $a = -\omega^2 x = 0$

35. **(2)**

Since the two waves are identical, they have same amplitudes.

The amplitude of the resultant wave is

$A = 2a\cos\frac{\phi}{2}$

Here, a = 10 mm, $\phi = 120°$

$A = 2\ (10\ \text{mm})\cos\frac{120°}{2}$

$= 2 \times 10 \times \frac{1}{2}\text{mm} = 10\text{mm}$

36. **(4)**

The given transverse harmonic wave equations is

$y = 3\sin\left(36t + 0.018x + \frac{\pi}{4}\right)$ ….. (i)

As there is positive sign between t and x terms therefore the given wave is travelling in the negative x directions. The standard transverse harmonic wave equations is

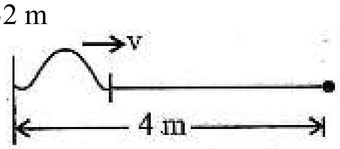
$y = a\sin(\omega t + kx + \varphi)$ ….. (ii)

Comparing (i) and (ii) we get
$a = 3cm, \omega = 36rads^{-1}, k = 0.018radcm^{-1}$
∴Amplitude of wave, a = 3 cm
Frequency of the wave,
$\upsilon = \frac{\omega}{2\pi} = \frac{36}{2\pi} = \frac{18}{\pi}$ Hz
Velocity of the wave
$v = \frac{\omega}{k} = \frac{36rads^{-1}}{0.018radcm^{-1}} = 2000cms^{-1}$
$= 20ms^{-1}$

37. **(2)**
Amplitude of reflected wave
$= \frac{2}{3} \times 0.6 = 0.4$
On reflection from a denser medium there is phase change of π.
∴Equations of reflected wave is
$y = 0.4 \sin 2\pi\left(t + \frac{x}{2} + \pi\right)$
$= -0.4 \sin 2\pi\left(t + \frac{x}{2}\right)$

38. **(2)**
For closed organ pipe,
$f = f_2 = 3f_1$
$= \frac{3 \times V}{4L} = \frac{3}{4L} \times \sqrt{\frac{\gamma p}{\rho_1}}$
For open organ pipe,
$f = f_2 = 2f_1 = \frac{2v}{2L'} = \frac{V}{L}$
$= \frac{1}{L'} \times \sqrt{\frac{\gamma p}{\rho_2}}$
For same sound frequency,
$\frac{3}{4L} \times \sqrt{\frac{\gamma P}{\rho_1}} = \frac{1}{L'} \times \sqrt{\frac{\gamma p}{\rho_2}}$
$L' = \frac{4L}{3} \times \sqrt{\frac{\rho_1}{\rho_2}}$

39. **(1)**
Distance travelled by the wave in four round trips 32 m



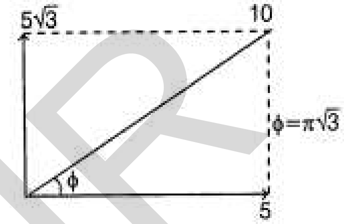
$t = \frac{d}{v} \Rightarrow v = \frac{d}{t} = \frac{32}{0.8} = 40 \text{ m/s}$

$v = \sqrt{\frac{T}{\mu}}$
$\Rightarrow T = \mu v^2$
$T = \frac{0.2}{4} \times (40)^2 = \frac{2 \times 16 \times 10}{4}$
$T = 80$ N

40. **(1)**

41. **(1)**

42. **(3)**
Given, $y_1 = 10\sin\left(3\pi t + \frac{\pi}{3}\right)$
$y_2 = 5(\sin 5\pi t + \sqrt{3}\cos 5\pi t)$



This can also be written as, $y_2 = 10\sin\left(5\pi t + \frac{\pi}{3}\right)$
Now, $A_1 = 10$ and $A_2 = 10$
$\therefore \frac{A_1}{A_2} = \frac{1}{1}$

43. **(4)**
Path difference $(\Delta x) = 50 \text{ cm} = \frac{1}{2}$ m
∴ Phase difference, $\Delta\phi = \frac{2\pi}{\lambda} \times \Delta x$
$\Rightarrow \phi = \frac{2\pi}{1} \times \frac{1}{2} = \pi$
Total phase difference $= \pi - \frac{\pi}{3} = \frac{2\pi}{3}$
$\Rightarrow A = \sqrt{a^2 + a^2 + 2a^2\cos\left(\frac{2\pi}{3}\right)} = a$

44. **(4)**
Time interval between two successive beats
$T = \frac{1}{\text{Beat frequency}} = \frac{1}{f_1 - f_2}$

45. **(3)**
Speed of sound wave in a medium $v \propto \sqrt{T}$ (where T is temperature of the medium). Clearly, when temperature changes speed also changes.
As, $v = \nu\lambda$
where ν is frequency and λ is wavelength. Frequency (ν) remains fixed
$\Rightarrow v \propto \lambda$ or $\lambda \propto v$
As frequency does not change, so wavelength (λ) changes.

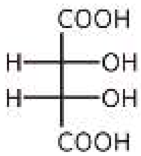
CHEMISTRY

46. **(4)**

Sol: Acidic strength ∝ –I effect ∝ – M effect

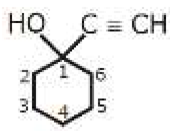
47. **(1)**

Sol:



P.O.S (Plane of Symmetry) Present

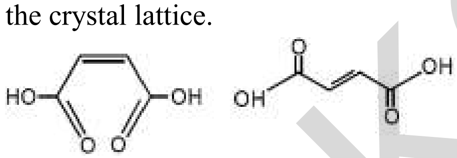
48. **(3)**



Sol: 1-Ethynyl cyclohexanol.

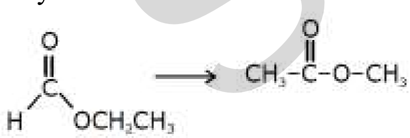
49. **(1)**

Sol: The melting point of fumaric acid is higher than that of maleic acid because the molecule of fumaric acid are more symmetric than those of maleic acid and hence, it get closely arranged in the crystal lattice.

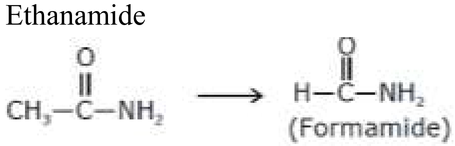


50. **(2)**

Sol: (A) Ethyl formate

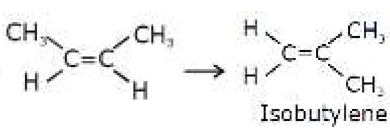


(B) Ethanamide



(C) Propanenitrile

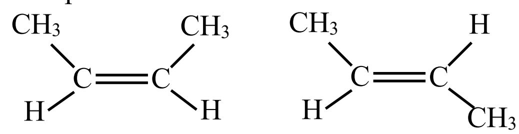
(D) 2-Butene



51. **(1)**

Sol: For G.I, the terminal Valencies should be different

∴ Option 1:

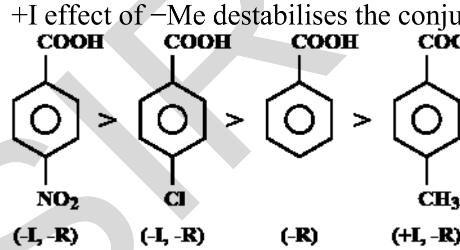


(Cis) (trans)

52. (1)

Sol: The order of acidity is correct and hence, assertion is correct.

This is because the -I and -R effect of $-NO_2$ and $-Cl$ group stabilises the conjugate base whereas +I effect of −Me destabilises the conjugate base.



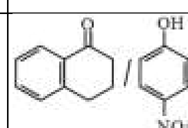
Hence, Reason is correct.

53. **(2)**

Sol: –Cl can also give +M effect, which increases electron density at ortho and para position.

54. **(4)**

Sol:

List-I (Mixture of compound)		List-II (Separation Technique)	
A.	H_2O/CH_2Cl_2	II.	Differential solvent extraction $CH_2Cl_2 > H_2O$ (density) so they can be separated by differential solvent extraction.
B.		III.	Column chromatography Due to H bond in p-nitro phenol.
C.	Kerosene/ Naphthalene	IV.	Fractional Distillation due to difference of b.pt between Kerosene/Naphthalene.
D.	$C_6H_{12}O_6/NaCl$	I.	Crystallization because NaCl is ionic.

Thus, correct match is A-II, B-III, C-IV, D-I

55. **(3)**

Sol: Sulphanilic acid does not show esterification test.
Presence of both sulphur and nitrogen give red colour in Lassigne's test.

56. **(1)**

Sol. Resonance energy $\propto$ Aromatic stability

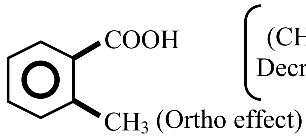
57. **(3)**

Sol.

$$\underset{\text{Aromatic}}{\text{(II)}} > \underset{\text{Conjugated}}{\text{(I)}} > \underset{\text{Non conjugated}}{\text{(III)}}$$

58. **(3)**

Sol.



$\left(\begin{array}{c}(CH_3 \rightarrow \text{EDG}) \\ \text{Decrease acidity}\end{array}\right)$

59. (2)

Sol.

60. (4)

Sol:

61. (4)

Sol.

62. (2)

Sol: Due to double bond character in $C_1 - C_2$

63. (2)

Sol. O^- follow +M effect in increase e^- density.

64. **(3)**

Sol. Wurtz Reaction:

$- 2CH_3CH_2Cl + 2Na \xrightarrow{\text{Ether}} CH_3 - CH_2 - CH_2 - CH_3 + 2NaCl$

65. **(2)**

Sol. Boiling α Mole cular mass α $\dfrac{1}{\text{number of branches}}$.

66. **(2)**

Sol. When butane-2-ol is heated with concentrated H_2SO_4, it undergoes dehydration to form an alkene. The major product formed is But-2-ene (But-2-en) due to the formation of a more stable internal alkene compared to But-1-ene.

67. **(4)**

Sol. $CH_2 = CH_2 + Br_2 \rightarrow BrCH_2 - CH_2 + Br$ (Brown) (Colour less) Disappearance of brown colour is the test for unsaturation

68. **(3)**

Sol. Stronger acid displaces weak acid from its salt.

69. **(1)**

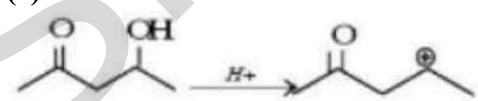
Sol. Anti-addition take place

70. **(3)**

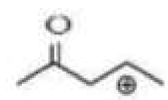
Sol. $R - C \equiv CH + \overset{\oplus}{H} \rightarrow \underset{\text{vinyl carbocation}}{R - \overset{\oplus}{C} = CH_2}$

71. **(2)**

Sol.



Is lees stable than

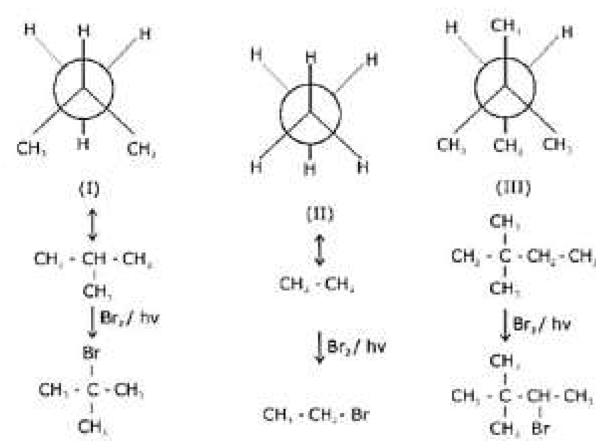


Due to resonance

Formed from (1) compound. The other two don't undergo dehydration.

72. **(1)**

Sol.

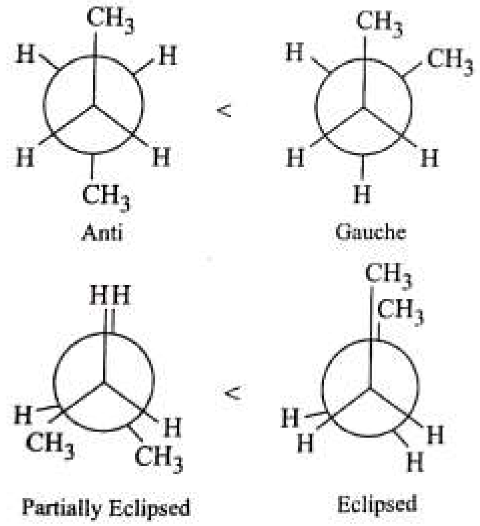


[Most stable free radical is formed as intermediate]

Order of Bromination = I > III > II

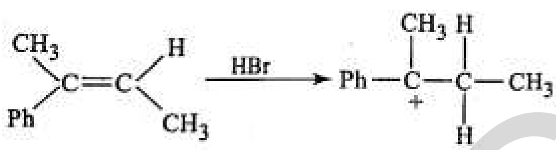
73. **(1)**

Sol. Increasing order of potential energy: I < III < IV < II



74. **(3)**

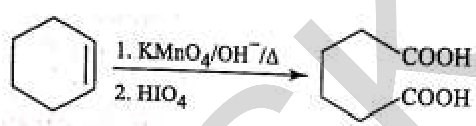
Sol.



Most Stable Carbocation

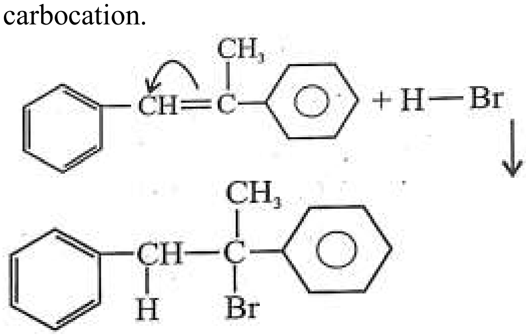
75. **(2)**

Sol.



76. **(2)**

Sol. The reaction is an example of electrophillic addition reaction & the intermediate here is a carbocation.



77. **(3)**

Sol. Sodium ethanoate is CH_3COONa and given process is soda-lime decarboxylation.

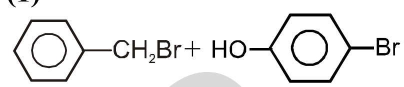
$$CH_3COONa + NaOH \xrightarrow{CaO} CH_4 + Na_2CO_3$$

Methane is obtained having molar mass 16. Two moles would be 32 g.

78. **(2)**

Sol. Due to + M effect of – OH group and hyperconjugation of – CH_3 group

79. **(1)**

Sol. 

80. **(2)**

Sol.

81. **(3)**

Sol.

82. **(3)**

Sol. 12th NCERT

83. **(3)**

Sol. Stability $C^+ \propto$ Reso > +M > +H > +I

84. **(2)**

Sol. More reso more stable C^+

85. **(1)**

Sol. Huckle rule

86. **(1)**

Sol. + M → – charge | Lone Pair on linked atom

87. **(1)**

Sol.

88. **(1)**

Sol. Non polar > octet > more reso > less reso

89. **(4)**

Sol. due to +H and +I effect

90. **(4)**

Sol. NH_2 has higher +M effect